Differential Splicing from Equivalence-Class Counts

Overview

Questions and unit of analysis

The analysis asks two distinct questions:

1. Does local splice-path usage differ by experimental condition within a cell type?

2. Does local splice-path usage differ across cell types within the same mice?

The primary analysis aggregates cells into *mouse-by-condition-by-cell-type pseudobulks* and models their observed paired-primer equivalence-class (EC) counts directly. Biological replication is represented by mouse random effects; cells are never treated as independent condition replicates.

A splice block is a local region between constitutive exonic anchors. Transcripts that take the same path through that region are collapsed to one subisoform. This targets local splicing choices, which can be identifiable even when the corresponding full-length isoforms are not.

For condition effects, each hypothesis is indexed by a splice block and a cell type. It is an omnibus test of

$$H_0: \quad \text{mean path usage is equal across the included conditions.}$$

Only conditions represented by at least three mouse pseudobulks are included. With two conditions and two paths, this is a precision-weighted differential-PSI test. With more conditions or paths, all condition coefficients and path log-ratios are tested jointly.

For cell-type effects, each hypothesis is indexed by a nonredundant splice-path partition. It is an omnibus test of equal path usage across the included cell types while condition is an unrestricted nuisance fixed effect and mouse is a random effect. Cell types must be represented in at least three mice. Both contrasts require at least 10 modeled gene UMIs per retained pseudobulk. The cell-type and condition hypothesis families are reported separately.

Workflow

The primary analysis proceeds in five steps:

1. Construct local splice blocks from the isoform annotation and map every supported isoform to a local path.

2. Aggregate paired oligo(dT) and random-hexamer EC counts within each mouse, condition, and cell type while retaining separate primer maps.

3. For each block, construct nested fixed-effect tensors that absorb all isoform directions orthogonal to the tested path contrast.

4. Fit multinomial and logistic-normal multinomial EC GLMMs with full-covariance variational posteriors, warm-starting each alternative from its fitted null.

5. Calibrate the nested objective gain by null simulation, cross-fitted tail modeling, and a global leave-block-out empirical null CDF; then apply BH separately to cell-type and condition hypotheses.

Path-estimate Gaussian, compositional, Dirichlet–multinomial, GEE, Laplace, cell-resolved, and joint-test analyses are supporting comparisons and appear in the appendices. They are not separate versions of the primary test.

Notation and Dimensions

An *equivalence class* (EC) is a set of isoforms compatible with one UMI. An EC count is therefore observed, whereas its allocation among compatible isoforms is latent. A *splice path* is an ordered local exon chain between two constitutive anchors. A *subisoform* is the set of full-length isoforms taking the same path through one block. PSI means percent spliced in; for two paths it is one component of the path-composition vector. ILR means isometric log-ratio, an orthonormal coordinate system for a composition.

Let $g = g(b)$ denote the gene containing block b, and let $\mathcal{I}_g$ be the set of isoforms retained by the paired-primer model and assigned to gene g by the isoform-to-gene table. Define $T_g = |\mathcal{I}_g| \in \mathbb{N}$. This is a gene-level set, not the set of isoforms whose exons overlap the block, and it is therefore shared by every block of the gene. Let $\mathcal{I}_b^{\text{sp}} \subseteq \mathcal{I}_g$ be the retained, annotated spliced isoforms represented in the block's GTF-derived path map. Isoforms in $\mathcal{I}_g \setminus \mathcal{I}_b^{\text{sp}}$, including modeled unspliced precursors, remain nuisance isoforms in the EC likelihood.

Let $I_b, S_b \in \mathbb{N}$ be the numbers of qualifying pseudobulks and represented spliced paths for block b, with $S_b \geq 2$. Then $d_b = S_b - 1 \in \mathbb{N}$ is its number of independent path contrasts. Define $r_g = T_g - 1$. There are $P = 2$ primer protocols, and primer p has $K_{gp} \in \mathbb{N}$ ECs for gene g. The principal symbols in the primary model are:

symbol	dimension or domain	meaning
b	$\{1, \ldots, \mathcal{B}\}$	splice-block index
i	$\{1, \ldots, I_b\}$	pseudobulk index within block b
p	$\{1, \ldots, P\}$	primer-protocol index
t	$\{1, \ldots, T_g\}$	isoform index within gene $g(b)$
s	$\{1, \ldots, S_b\}$	splice-path index
$\boldsymbol{c}_{gip}$	$\mathbb{N}_0^{K_{gp}}$	observed pseudobulk EC counts
A_{gp}	$\mathbb{R}_+^{K_{gp} \times T_g}$	weighted EC-by-isoform map
$\boldsymbol{\eta}_{gi,-}$	$\mathbb{R}^{r_g}$	free isoform logits
$\boldsymbol{q}_{gip}$	$\Delta^{K_{gp}-1}$	primer-specific EC probabilities
Q_b	$\mathbb{R}^{S_b \times d_b}$	Helmert ILR basis
H_b	$\mathbb{R}^{r_g \times d_b}$	tested block-path basis
N_b	$\mathbb{R}^{r_g \times (T_g - S_b)}$	orthogonal nuisance basis
$\boldsymbol{u}_{gm}$	$\mathbb{R}^{r_g}$	mouse random effect
$\boldsymbol{e}_{gi}$	$\mathbb{R}^{r_g}$	logistic-normal pseudobulk effect

Here $\mathcal{B} \in \mathbb{N}$ is the number of blocks, $\mathbb{N}_0 = \{0, 1, \ldots\}$, $\Delta^{S_b-1} = \{\boldsymbol{v} \in \mathbb{R}_+^{S_b} : \boldsymbol{1}'\boldsymbol{v} = 1\}$, and $\mathbb{S}_+^k$ is the set of symmetric positive-semidefinite $k \times k$ matrices. Bold lower-case symbols are vectors, upper-case symbols are matrices, $\mathbf{1}_k \in \mathbb{R}^k$ is a vector of ones, $\mathbf{0}_k \in \mathbb{R}^k$ is a vector of zeros, $I_k \in \mathbb{R}^{k\times k}$ is the identity, $\odot$ is elementwise multiplication, and $(\cdot)^+$ is the Moore–Penrose pseudoinverse.

Helmert basis and ILR coordinates

The simplex has only $d_b = S_b - 1$ free directions because path proportions sum to one. The Helmert matrix Q_b is a fixed orthonormal basis for the sum-zero subspace:

$$Q_b' Q_b = I_{d_b}, \qquad Q_b' \mathbf{1}_{S_b} = \mathbf{0}_{d_b}.$$

Using one-based row $r \in \{1, \ldots, S_b\}$ and column $j \in \{1, \ldots, d_b\}$, its entries are

$$(Q_b)_{rj} = \begin{cases} 1/\sqrt{j(j+1)}, & r \leq j, \\ -j/\sqrt{j(j+1)}, & r = j+1, \\ 0, & r > j+1. \end{cases}$$

For any strictly positive $\boldsymbol{\psi}_{bi} \in \Delta^{S_b-1}$, the ILR coordinate is $\boldsymbol{y}_{bi} = Q_b' \log \boldsymbol{\psi}_{bi} \in \mathbb{R}^{d_b}$, where the logarithm is elementwise. For $S_b = 2$, $Q_b = (1, -1)'/\sqrt{2}$ and $y_{bi} = \log(\psi_{bi1}/\psi_{bi2})/\sqrt{2}$. Thus a two-path ILR test is a scaled log-odds, or differential-PSI, test.

Primary EC-count GLMMs

Observed data and primer-specific EC probabilities

Fix a gene g and one of its splice blocks b. Let $T_g \in \mathbb{N}$ be the number of EC-supported isoforms and $r_g = T_g - 1$ the number of free reference-logit coordinates. There are $I_b \in \mathbb{N}$ retained pseudobulks and $M_b \in \mathbb{N}$ represented mice. For primer protocol $p \in \{1, 2\}$, let $K_{gp} \in \mathbb{N}$ be the number of retained ECs,

$$\boldsymbol{c}_{gip} \in \mathbb{N}_0^{K_{gp}}, \qquad n_{gip} = \mathbf{1}_{K_{gp}}' \boldsymbol{c}_{gip} \in \mathbb{N}_0, \qquad A_{gp} \in \mathbb{R}_+^{K_{gp}\times T_g}.$$

Here A_{gpkt} is the fixed compatibility or conditional weight of EC k for isoform t. Oligo(dT) and random-hexamer observations have separate count vectors and A_{gp} matrices, so primer-specific positional coverage enters the likelihood without changing the underlying isoform composition. Let $\boldsymbol{\eta}_{gi} = (\boldsymbol{\eta}_{gi,-}', 0)' \in \mathbb{R}^{T_g}$ append a zero reference-isoform logit and define

$$q_{gipk}(\boldsymbol{\eta}_{gi}) = \frac{\sum_{t=1}^{T_g} A_{gpkt} \exp(\eta_{git})}{\sum_{h=1}^{K_{gp}} \sum_{t=1}^{T_g} A_{gpht} \exp(\eta_{git})}, \qquad \boldsymbol{q}_{gip} \in \Delta^{K_{gp}-1}.$$

Conditioning on each observed primer total removes library size from the composition likelihood. An EC may be compatible with multiple isoforms; there is no preliminary allocation of its count to a transcript or path. Unsupported isoforms and ECs not assigned to g are excluded, but every supported isoform of g, including an unspliced nuisance isoform, remains in every block likelihood for that gene.

Multinomial EC GLMM

Let $m(i) \in \{1, \ldots, M_b\}$ denote the mouse for pseudobulk i. The multinomial EC GLMM is,

$$\boldsymbol{c}_{gip} \mid \boldsymbol{\eta}_{gi} \sim \text{Multinomial}\{n_{gip}, \boldsymbol{q}_{gip}(\boldsymbol{\eta}_{gi})\}, \tag{1}$$

$$\boldsymbol{\eta}_{gi,-} = \mathcal{F}_{bi}(\boldsymbol{\beta}_b) + \boldsymbol{u}_{g,m(i)}, \tag{2}$$

$$\boldsymbol{u}_{gm} \sim N(\mathbf{0}_{r_g}, \sigma_g^2 I_{r_g}), \qquad \boldsymbol{u}_{gm} \in \mathbb{R}^{r_g}, \quad \sigma_g > 0. \tag{3}$$

The same mouse effect enters both primer halves and every pseudobulk from that mouse. The fixed predictor $\mathcal{F}_{bi}(\boldsymbol{\beta}_b) \in \mathbb{R}^{r_g}$ is a tensor design defined below. Thus the model uses observed EC counts, preserves gene-wide ambiguity, and tests a local block contrast without first estimating a response vector.

Logistic-normal EC GLMM

The logistic-normal extension adds one isotropic residual composition effect per pseudobulk,

$$\boldsymbol{\eta}_{gi,-} = \mathcal{F}_{bi}(\boldsymbol{\beta}_b) + \boldsymbol{u}_{g,m(i)} + \boldsymbol{e}_{gi}, \tag{4}$$

$$\boldsymbol{e}_{gi} \sim N(\mathbf{0}_{r_g}, \tau_g^2 I_{r_g}), \qquad \boldsymbol{e}_{gi} \in \mathbb{R}^{r_g}, \quad \tau_g > 0. \tag{5}$$

The multinomial observation model is otherwise unchanged. The same $\boldsymbol{e}_{gi}$ enters the oligo(dT) and random-hexamer likelihood terms because it perturbs the latent isoform composition rather than primer-specific EC probabilities. It therefore represents correlated extra-multinomial variation across the paired primer halves. The multinomial model is the special case $\tau_g = 0$. This logistic-normal construction is the primary overdispersion model; the direct EC Dirichlet–multinomial is retained only as an appendix sensitivity analysis.

Block contrast and nested fixed-effect tensors

Let $\pi_b(t) \in \{-1, 1, \ldots, S_b\}$ be isoform t's represented path, where -1 marks an isoform retained only as a likelihood nuisance. Lift the Helmert basis $Q_b \in \mathbb{R}^{S_b \times d_b}$ from paths to isoforms by defining $\widetilde{D}_b \in \mathbb{R}^{T_g \times d_b}$ rowwise,

$$(\widetilde{D}_b)_{t\cdot} = \begin{cases} (Q_b)_{\pi_b(t)\cdot}, & \pi_b(t) \geq 1, \\ \mathbf{0}'_{d_b}, & \pi_b(t) = -1. \end{cases}$$

Thus isoforms taking the same represented path receive the same row, whereas nuisance isoforms receive zero. Center each column over isoforms using $C_g = I_{T_g} - \mathbf{1}_{T_g}\mathbf{1}'_{T_g}/T_g$ and define $D_b = C_g\widetilde{D}_b \in \mathbb{R}^{T_g \times d_b}$. Let $R_g = [I_{r_g} \ \ -\mathbf{1}_{r_g}] \in \mathbb{R}^{r_g \times T_g}$ convert full logits to logits relative to isoform T_g. The tested basis is

$$H_b = R_g D_b = D_{b,1:r_g,:} - \mathbf{1}_{r_g} D_{b,T_g,:} \in \mathbb{R}^{r_g \times d_b}.$$

Because $R_g \mathbf{1}_{T_g} = \mathbf{0}_{r_g}$, the column centering cancels from H_b. It fixes a convenient sum-zero representative in full-logit space but does not change any reference-logit contrast.

Let $N_b^{\text{full}} \in \mathbb{R}^{T_g \times (T_g - S_b)}$ be an orthonormal basis for the full-logit subspace orthogonal to both D_b and $\mathbf{1}_{T_g}$, and set $N_b = R_g N_b^{\text{full}} \in \mathbb{R}^{r_g \times (T_g - S_b)}$. Orthogonality is therefore imposed before reference conversion. The columns of $[N_b \ H_b] \in \mathbb{R}^{r_g \times r_g}$ span every free isoform-logit direction. Consequently, nuisance effects in N_b cannot express the tested local path contrast, while the alternative remains unrestricted after the H_b columns are added.

This construction separates the biological comparison from the local splicing choice. The columns of H_b encode the d_b independent between-path log-ratios, while the columns of W_b identify the cell-type or condition differences being tested. For a two-path block, $Q_b = (1, -1)'/\sqrt{2}$. Before reference conversion, mapped isoforms on the first path receive $+1/\sqrt{2}$, mapped isoforms on the second receive $-1/\sqrt{2}$, and nuisance isoforms receive zero. The resulting single column of H_b is the same direction expressed relative to the chosen reference isoform. The nuisance basis N_b still permits factor-associated changes within paths and in other gene-level isoform directions. Thus the null does not require equal isoform composition across groups. It excludes only group differences that project onto the block's local path balance, and the alternative restores those directions through $H_b\boldsymbol{\gamma}_{bj}$.

Let $Z_b \in \mathbb{R}^{I_b \times q_0}$ be the nuisance design and $W_b \in \mathbb{R}^{I_b \times q_1}$ the treatment-coded tested design. The nested predictors are

$$\mathcal{F}_{bi}^{(0)} = B_0' Z_{bi}' + \sum_{j=1}^{q_1} W_{bij} N_b \boldsymbol{\alpha}_{bj}, \tag{6}$$

$$\mathcal{F}_{bi}^{(1)} = \mathcal{F}_{bi}^{(0)} + \sum_{j=1}^{q_1} W_{bij} H_b \boldsymbol{\gamma}_{bj}, \tag{7}$$

where $B_0 \in \mathbb{R}^{q_0 \times r_g}$, $\boldsymbol{\alpha}_{bj} \in \mathbb{R}^{T_g - S_b}$, and $\boldsymbol{\gamma}_{bj} \in \mathbb{R}^{d_b}$. The null sets every $\boldsymbol{\gamma}_{bj} = \mathbf{0}_{d_b}$; the alternative adds $q_1 d_b$ block-path coefficients. In implementation these expressions are stored as tensors $\mathcal{X}_b^{(h)} \in \mathbb{R}^{I_b \times r_g \times a_h}$, $h \in \{0, 1\}$, and evaluated as $\mathcal{F}_{bi}^{(h)} = \mathcal{X}_{bi}^{(h)} \boldsymbol{\beta}_b^{(h)}$.

The two biological tests differ only in Z_b and W_b,

- For the cell-type test, Z_b is a one-hot condition design and W_b has $J_b - 1$ columns, where J_b is the number of cell types. The tested dimension is $(J_b - 1)d_b$. All condition-by-isoform effects and cell-type effects orthogonal to the block contrast are present under the null.
- For condition within cell type, the data are restricted to one cell type, $Z_b = \mathbf{1}_{I_b}$, and W_b has $C_b^{\text{cond}} - 1$ condition columns. The tested dimension is $(C_b^{\text{cond}} - 1)d_b$. Each retained condition must be represented by at least three mice.

Distinct annotated blocks can induce the same partition after unsupported isoforms are removed. Before fitting or BH correction, tests are therefore deduplicated by gene, supported isoform set, canonical path partition, covered pseudobulks, and tested factor levels.

Full-covariance variational fitting

For mouse m, let $h_{gm} \in \mathbb{N}$ be its number of retained pseudobulks. In the multinomial model define $\boldsymbol{z}_{gm} = \boldsymbol{u}_{gm} \in \mathbb{R}^{r_g}$. In the logistic-normal model,

$$\boldsymbol{z}_{gm} = (\boldsymbol{u}_{gm}', \boldsymbol{e}_{gi_1}', \ldots, \boldsymbol{e}_{gi_{h_{gm}}}')' \in \mathbb{R}^{L_{gm}}, \qquad L_{gm} = (1 + h_{gm}) r_g.$$

The prior covariance is,

$$P_{gm} = \operatorname{diag}(\sigma_g^2 I_{r_g}, \tau_g^2 I_{h_{gm} r_g}) \in \mathbb{S}_{++}^{L_{gm}},$$

with the residual block omitted for the multinomial model. The variational family factorizes across mice but has unrestricted covariance within mouse,

$$Q(\boldsymbol{z}_g) = \prod_{m=1}^{M_b} N(\boldsymbol{z}_{gm}; \boldsymbol{\mu}_{gm}, \Sigma_{gm}), \qquad \Sigma_{gm} \in \mathbb{S}_{++}^{L_{gm}}.$$

Every Σ_{gm} is parameterized by a lower Cholesky factor with an exponentiated diagonal. It captures posterior dependence among isoform logits and, in the logistic-normal model, between the shared mouse effect and all residuals from that mouse. It does not add an unstructured covariance parameter to the generative model.

The multinomial likelihood uses the analytic tilted log-sum-exp bound. For a Gaussian logit vector with mean $\boldsymbol{m} \in \mathbb{R}^{T_g}$, covariance $V \in \mathbb{S}_+^{T_g}$, and local coordinate $\boldsymbol{a} \in \Delta^{T_g-1}$,

$$E \log \sum_t \exp(\eta_t) \leq \frac{1}{2}\boldsymbol{a}'V\boldsymbol{a} + \log \sum_t \exp\{m_t + \tfrac{1}{2}V_{tt} - (V\boldsymbol{a})_t\}, \tag{8}$$

$$\boldsymbol{a} = \operatorname{softmax}\{\boldsymbol{m} + \tfrac{1}{2}\operatorname{diag}(V) - V\boldsymbol{a}\}. \tag{9}$$

The local coordinate is iterated for eight fixed-point updates. At this accuracy, differentiating through the updates changes the objective gradient by less than 10^{-8} in the benchmark fits, so the implementation stops the local-coordinate gradient and uses the envelope theorem. The Gaussian KL term is exact,

$$\mathrm{KL}(Q_m \| P_m) = \frac{1}{2}\left[\operatorname{tr}(P_m^{-1}\Sigma_m) + \boldsymbol{\mu}_m' P_m^{-1}\boldsymbol{\mu}_m - L_{gm} + \log|P_m| - \log|\Sigma_m|\right].$$

Summing bounded expected log likelihoods and subtracting these KL terms gives the optimized objective $\mathcal{L}_b^{(h)} \in \mathbb{R}$. To improve conditioning, let $s_{gmj} > 0$ be the prior standard deviation of latent coordinate j, let $\widetilde{\mu}_{gmj} \in \mathbb{R}$ be its optimizer-scale mean, and let $\widetilde{L}_{gmjk} \in \mathbb{R}$ be element (j,k) of its optimizer-scale lower Cholesky factor. The physical variational parameters are obtained by the row scaling $\mu_{gmj} = s_{gmj}^{3/4}\widetilde{\mu}_{gmj}$ and $L_{gmjk} = s_{gmj}^{3/4}\widetilde{L}_{gmjk}$. This is an invertible reparameterization, not a change to the posterior family, and reduces the ridge between latent moments and their estimated prior scales.

JAX supplies 64-bit objective values and gradients; bounded L-BFGS runs for at most 300 strict iterations. A fit that reaches this limit resumes from the complete previous parameter vector for at most 50 iterations with a relaxed relative-objective tolerance. The returned parameters are transformed back to the common physical representation. The alternative copies the fitted null fixed effects, variational means, Cholesky factors, and variance parameters, initializes only the new block coefficients to zero, and then optimizes. This warm start keeps the nested objective comparison on the same local branch.

Bouchard coordinate-ascent alternative

A deterministic alternative uses Bouchard's quadratic softmax bound. EC-to-isoform responsibilities give a linear lower bound for each positive EC numerator, while the Bouchard–Jaakkola construction gives a quadratic upper bound for each subtracted primer normalizer. After collecting terms, pseudobulk i's bounded log likelihood has the form

$$-\tfrac{1}{2}E_q(\boldsymbol{\eta}_{gi,-}'\Omega_{gi}\boldsymbol{\eta}_{gi,-}) + \boldsymbol{h}_{gi}'E_q(\boldsymbol{\eta}_{gi,-}) + C_{gi}, \qquad \Omega_{gi} \in \mathbb{S}_+^{r_g}, \quad \boldsymbol{h}_{gi} \in \mathbb{R}^{r_g}.$$

This restores closed-form coordinate updates for every full-covariance mouse factor, the fixed effects, and the empirical-Bayes variance components. One sweep updates EC responsibilities, Bouchard location and curvature parameters, mouse Gaussian moments, fixed effects, and variance components. No gradient optimizer is used. Because the fixed effects and variances remain point estimates, this is coordinate-ascent variational EM; conjugate priors would give fully Bayesian CAVI. The complete derivation and dimensions are given in `docs/cavi.tex`.

The CAVI posterior used the same packed representation as the tilted fit and could initialize tilted L-BFGS directly. In six null and six effect random-intercept simulations, 25 CAVI initialization sweeps reduced median tilted iterations from 39.5 to 37.0 under the null and from 45.5 to 43.5 under the effect, but did not change convergence or the final objective and increased total time. In the harder three-isoform logistic-normal simulations, direct and hybrid tilted fitting each converged in four of six null replicates, while direct fitting converged in four of six effect replicates and the hybrid in three of six. Standalone CAVI had a much looser bound and larger coefficient error. The derivation and benchmark are retained as a negative result, but the CAVI fitter and initialization options were removed from the package and analysis runners.

The 65-gene real-data logistic-normal screens make the CAVI result stronger. The established Laplace-initialized tilted fit converged under both nested models for 52 condition genes and 56 cell-type genes, or 80.0% and 86.2%. With 25 CAVI initialization sweeps, the same tilted optimizer converged under both models for only 11 condition genes and 10 cell-type genes, and rescued none of the historical failures. Where both initializations converged, their final tilted objectives agreed closely. The Bouchard solution can lead to the same optimum, but its loose separable bound places the optimizer too far from that optimum to improve convergence in these data.

A contemporaneous rerun compared optimizer schedules on the same 65 genes per contrast. The original 25-update, fully differentiated schedule converged for both nested models in 40 genes for each contrast. Eight local updates, envelope gradients, three-quarter prior-scale reparameterization, and selective relaxed continuation increased joint convergence to 64 of 65 genes in each contrast. The remaining gene had an empty EC matrix for one primer; treating this likelihood contribution as its mathematical value of zero, rather than evaluating an indeterminate zero times negative infinity, made both nested fits converge in both contrasts and raised corrected joint convergence to 65 of 65. Median time per nested-model pair decreased from 6.47 to 5.86 seconds for cell type and from 6.76 to 5.97 seconds for condition. On matched genes, the reparameterized schedule had median objective improvements of 0.18–0.24 for three of the four contrast-by-model combinations and 0.015 for the fourth, with no objective more than 0.33 below the original fit. These results support optimizer conditioning rather than CAVI initialization as the convergence remedy.

Lower objective precision was tested on 24 difficult gene–contrast fits. fp32 reduced median time per nested pair from 11.38 to 8.58 seconds and reported convergence for every fit, but its null and alternative objectives were lower than fp64 by medians of 0.22 and 0.68. The nested-model objective gain changed by a median absolute 0.45 and by as much as 9.27. On a representative hard gene, bfloat16 gave coarse objective values, failed the null fit, and was slower than fp64. A 25-iteration fp32 warm-up followed by fp64 refinement restored objective accuracy and found one alternative optimum 7.52 units above the direct fp64 fit, but increased median time by 34%. It is useful as a sensitivity restart, not as the default initializer. Final fitting therefore remains fp64.

Algorithm 1 Fit one multinomial or logistic-normal block EC GLMM test

Require: Paired counts $\boldsymbol{c}_{gip}$, maps A_{gp}, block path map π_b, labels, mouse indices, and model type
Ensure: Nested-fit statistic T_b and fitted null for simulation

Retain pseudobulks with at least 10 modeled gene UMIs and enforce the three-mouse representation rules.
Construct H_b, N_b, and deduplicate the supported partition.
Build $Z_b, W_b, \mathcal{X}_b^{(0)}, \mathcal{X}_b^{(1)}$ for the chosen cell-type or condition contrast.
Initialize the null full-covariance posterior and variance parameters.
repeat
    Iterate every tilted local coordinate for eight updates and apply the envelope theorem.
    Evaluate the bounded ELBO and its 64-bit gradient in prior-scale optimizer coordinates.
    Update fixed effects, scaled posterior means and Cholesky rows, and $\log \sigma_g$ plus optional $\log \tau_g$ by bounded L-BFGS.
until convergence or the retry budget is exhausted
Expand the null parameter vector with zero block coefficients and fit the alternative from this warm start.
Set $T_b \leftarrow 2(\mathcal{L}_b^{(1)} - \mathcal{L}_b^{(0)})$.

Null simulation and resolved-tail calibration

For each block, the fitted null generates five parametric-bootstrap replicates. Each replicate draws mouse effects and, for the logistic-normal model, pseudobulk effects; forms primer-specific EC probabilities through A_{gp}; preserves every observed n_{gip}; draws multinomial EC counts; and refits both nested models with the same warm-start and stopping rules.

Raw empirical ranks within tested-dimension strata are calibrated at ordinary levels but do not resolve genome-wide tails: after deduplication their minimum p-values are approximately 4.5×10^{-4}, while BH requires values near 10^{-5} at the first ranks. Tail calibration therefore proceeds as follows. Within each likelihood and tested-dimension stratum, let u_s be the 90th percentile of bootstrap statistics. Excesses $T - u_s > 0$ follow a fitted generalized Pareto distribution with shape $\xi_s \in \mathbb{R}$ and scale $\beta_s > 0$. Entire block tests, including all their bootstrap replicates, are assigned to one of five folds. The tail model fitted on four folds maps statistics in the held-out fold to lower-tail scores $\rho \in (0, 1]$.

Residual GPD error is removed by a second empirical calibration. Let $\mathcal{R}_{-b}$ contain all cross-fitted null scores except scores generated for block b. The reported p-value is

$$p_b = \frac{1 + \sum_{\rho \in \mathcal{R}_{-b}} \mathbf{1}\{\rho \leq \rho_b\}}{1 + |\mathcal{R}_{-b}|}.$$

Every bootstrap replicate receives the same leave-block-out calibration. Aggregate null rejection is audited at 0.05, 0.01, and 0.001, and 0.05 rejection is also checked by gene-depth and sample-count quartile. BH values are released only when aggregate and quartile 0.05 rejection rates lie in the prespecified interval 0.025–0.075. Thresholds 80%, 85%, 90%, and 95% are reported as a tail-model sensitivity analysis; a robust call must pass BH in all four analyses.

Algorithm 2 Cross-fitted calibration and genome-wide correction

Require: Observed statistics, fitted nulls, five null replicates per test, and tail threshold q
Ensure: Calibrated p-values and BH values

```
for all tests b do
    Simulate paired-primer counts from the fitted null while preserving observed primer totals; refit both nested models.
end for
Pool converged null statistics by likelihood and tested dimension; combine dimensions represented by fewer than 100 tests into a rare stratum.
for f = 1, ..., 5 do
    Fit each GPD tail using blocks outside fold f.
    Score observed and null statistics belonging to fold f.
end for
Recalibrate every score against the global leave-block-out null-score CDF within likelihood.
Audit null rejection overall and by depth and sample count.
Apply BH separately to the cell-type and within-cell-type condition families.
```

Primary Results

At the 90% tail threshold, the nonredundant logistic-normal analysis tests 4,825 cell-type partitions and 14,833 block-by-cell-type condition hypotheses. Table 1 reports the results. The null rates at 0.05, 0.01, and 0.001 are from held-out block tests and therefore assess the complete tail-scoring and empirical-recalibration procedure.

Table 1: Primary logistic-normal EC GLMM results. Robust calls pass BH at every GPD threshold from 80% through 95%.

Contrast	Tests	Conv.	Nominal	FDR	Robust	Null < .05	Null < .01	Null < .001
Cell type	4,825	0.999	771	252	238	0.04999	0.00999	0.00099
Condition	14,833	1.000	1,397	0	0	0.05001	0.00999	0.00100

For cell type, 80%, 85%, 90%, and 95% GPD thresholds produce 242, 240, 252, and 245 calls; 238 occur in all four sets. The multinomial EC GLMM identifies 272 partitions at the 90% threshold and overlaps 231 of the 238 robust logistic-normal calls. The direct EC Dirichlet–multinomial appendix sensitivity identifies 59, with 45 partitions called by all three EC likelihoods. Thus the cell-type result is not specific to the logistic-normal variance model.

For condition within cell type, the same thresholds produce zero, 28, zero, and zero calls, with an empty intersection. At the primary 90% threshold the minimum FDR is 0.0545. The data therefore support strong cell-type DS but no condition DS event that is robust to the tail threshold.

Nonbootstrap Laplace sensitivity

Table 2 compares the analytic Laplace LRT and BIC Bayes-factor approximation with the completed mouse bootstrap analysis. A Laplace fit is retained only when the infinity norms of its scaled outer gradient and random-effect mode score are each at most 10^{-4}. Scaling the objective and gradient by the absolute initial objective changes neither the optimum nor the likelihood ratio, but prevents count depth from setting the numerical line-search scale.

Table 2: Nonbootstrap mouse cell-type sensitivity. BF counts use $\widehat{\mathrm{BF}}_{10}^{\mathrm{BIC}} > 10$; bootstrap calls use the primary GPD calibration.

Likelihood	Tests	Laplace conv.	LRT nominal	LRT BH	BF	Bootstrap BH
Multinomial	4,825	0.942	710	322	160	272
Logistic-normal	4,825	0.936	646	271	128	252

Restricting comparison to tests converged under both fitting strategies gives Laplace-versus-bootstrap statistic rank correlations 0.972 for multinomial and 0.964 for logistic-normal; p-value rank correlations are 0.937 and 0.922. The multinomial analyses share 247 BH calls, and the logistic-normal analyses share 204. Among these matched tests, the BF threshold selects 160 and 128 events, of which 142 and 116 are bootstrap BH calls. Applying the nominal chi-square reference to the existing variational-fit null simulations gives 0.05 rejection rates 0.0389 and 0.0374, indicating mild conservatism for that diagnostic rather than inflation.

The logistic-normal alternative estimates its observation-noise standard deviation at the lower numerical bound in 758 of 4,438 jointly converged fits. Because this nuisance variance is then on a boundary, its Wilks reference is less regular. The multinomial LRT is the cleaner nonbootstrap sensitivity, while the BF threshold supplies a smaller high-evidence subset; neither supersedes the empirically calibrated primary result without the mouse comparison above.

The independent human analysis gives the results in Table 3. The recommended multinomial likelihood was rerun with 12 posterior-mode Newton steps and implicit mode differentiation on the first attempt, with 30 steps on continuation attempts; the logistic-normal row is the previous supporting sensitivity. The condition calls span 137 distinct blocks and 135 genes; 176 of 193 also exceed the BIC BF threshold. EX1, EX2, EX5, EX3, and EX4 contribute 62, 34, 21, 19, and 14 condition calls, respectively, with the remainder distributed across other neuronal and glial cell types.

Table 3: Nonbootstrap human EC-block results. Condition hypotheses are block-by-cell-type tests. BF counts include only converged nested fits.

Contrast	Likelihood	Tests	Conv.	Nominal	LRT BH	BF
Cell type	Multinomial	11,597	0.947	2,557	1,463	239
Cell type	Logistic-normal	11,597	0.943	2,161	1,125	148
Condition	Multinomial	73,144	0.940	5,497	193	522

The condition nominal rate is 0.0799 among converged tests, whereas the cell-type multinomial rate is 0.2331, as expected for the stronger biological contrast. The BF and BH counts need not be ordered: BF > 10 is an uncorrected evidence threshold whose BIC penalty depends on tested dimension and subject count, while BH controls a genome-wide frequentist error rate.

Matched external-method comparison

LeafCutter, scQuint, and MAJIQ are compared on junction evidence from the same cells and biological-replicate definitions used by the EC GLMM. Reads are aligned with STARsolo using the three-round Split-seq barcode geometry. Spliced alignments are restricted to unique genomic mappings and deduplicated by cell barcode, UMI, genomic start, and CIGAR. Junction UMIs are then summed within subject-by-cell-type pseudobulks. This prevents individual cells from being treated as independent biological replicates and gives every external method the same subject and cell-type support.

Condition tests compare conditions separately within each cell type. Cell-type tests use only subjects represented in both cell types. LeafCutter includes subject as a covariate for the paired cell-type tests. scQuint is run on the pseudobulk junction matrix because its differential-splicing interface has no covariate argument. MAJIQ v3 builds one common splice graph and PSI-coverage object from the pseudobulk BAMs, then applies its replicate-level Heterogen Mann–Whitney test to each LSV edge. Its raw test p-value is used for BH. The 95th percentile over tests of posterior PSI draws is retained as a sensitivity statistic, but MAJIQ documents that quantity as uncalibrated as a p-value, so it is not used for MHT. MAJIQ and scQuint therefore use the matched replicates but cannot model their cell-type pairing. Each method receives BH correction within a pairwise contrast. For a cell-type-wide feature score, pairwise p-values are combined by Simes before a second BH correction. The comparison is run unchanged on both the mouse microglialess data and an independent human Parse Biosciences brain dataset.

Two MAJIQ coverage configurations are retained. The stringent configuration uses its bulk-RNA defaults: 50% experiment support at splice-graph build and Heterogen, then at least 10 reads across three nonzero bins per pseudobulk for PSI coverage. Because all 936 mouse pseudobulks form one build group, the first threshold requires global support in approximately 468 pseudobulks. The relaxed UMI configuration instead requires absolute support in three pseudobulks at build, three UMIs in one nonzero bin for PSI coverage, and three quantified pseudobulks in each Heterogen arm. It retains MAJIQ's strict, nonredundant LSV selection.

Tables 4 and 5 report the resulting genome-wide test and discovery counts. These are comparisons of practical discovery yield, not estimates of relative power: no ground truth is available, and the methods test different units. Tealeaf tests EC-supported splice-block path partitions, LeafCutter tests intron clusters, MAJIQ tests edges within local splicing variations (LSVs), and scQuint tests intron groups. Their filters consequently produce different testing universes. The discovery fraction is reported to make that difference visible, but it is not an accuracy measure. In particular, a larger fraction can reflect stronger filtering as well as greater sensitivity.

Table 4: Matched mouse differential-splicing results. Tealeaf uses the ordinary multinomial EC GLMM with the same optimized Laplace LRT used for the human data; the separately reported calibrated logistic-normal analysis remains the primary mouse validation result. External-method cell-type entries are Simes omnibus tests over pairwise contrasts; their condition entries are pairwise feature-by-contrast tests. All discovery counts use BH FDR 0.05. Rate is discoveries divided by tests.

		Cell type			Condition		
Method	Test unit	Tests	BH	Rate	Tests	BH	Rate
Tealeaf EC GLMM	EC block	4,825	322	6.67%	14,833	3	0.02%
LeafCutter	Intron cluster	1,020	229	22.45%	10,279	40	0.39%
MAJIQ Heterogen	LSV edge	46	27	58.70%	3,836	34	0.89%
scQuint	Intron group	16,081	828	5.15%	161,711	16	0.01%

Table 5: Matched human differential-splicing results. Tealeaf uses the ordinary multinomial EC GLMM with the same optimized Laplace LRT used for the mouse data; external-method summaries are constructed as in Table 4. All discovery counts use BH FDR 0.05. Rate is discoveries divided by tests.

		Cell type			Condition		
Method	Test unit	Tests	BH	Rate	Tests	BH	Rate
Tealeaf EC GLMM	EC block	11,597	1,463	12.62%	73,144	193	0.26%
LeafCutter	Intron cluster	2,155	238	11.04%	5,238	475	9.07%
MAJIQ Heterogen	LSV edge	582	342	58.76%	3,522	278	7.89%
scQuint	Intron group	27,097	2,239	8.26%	85,852	36	0.04%

Table 8 compares aggregate worker time and peak resident memory for the broad cell-type and condition campaigns that produced Tables 4 and 5. Aggregate worker time is the sum of elapsed time over analysis tasks, so it is independent of queue delay and parallelism. Peak memory is the maximum for one worker. The accounting starts from each method's prepared analysis input and includes model fitting and method-specific result construction, but excludes shared read processing. Tealeaf uses the same ordinary multinomial EC GLMM and Laplace LRT in both datasets. Each first attempt uses 12 posterior-mode Newton updates and implicit differentiation, failed-convergence fits receive a 30-update continuation, and compiled objectives are reused within bounded gene-and-shape groups. On a complete 72-test human condition shard, implicit differentiation reduced elapsed time from 79.7 to 37.1 minutes and peak memory from 6.88 to 6.66 GiB relative to differentiating through the Newton updates, with the same 69 jointly converged tests and median absolute statistic difference below 10^{-12}. Gene-group compilation reuse reduced a second shard benchmark from 37.9 to 37.1 minutes. These pre-analytic broad reruns used 118.8 mouse worker hours and 4.74 GiB peak worker memory, and 729.4 human worker hours and 6.83 GiB; they were 7.8-fold slower than scQuint in each dataset. The condition families were not rerun after the exact derivative change, so Table 6 supersedes these resource totals only for the matched cell-type analysis. The human run has 4.3-fold more tests than mouse and its median test has five rather than three isoforms, 26 rather than nine ECs, and five rather than two block paths; these larger likelihoods account for most of the remaining dataset difference. Lower-precision and early-stopping experiments did not preserve accuracy or reduce real-data elapsed time, and CAVI initialization reduced convergence, so none was used for the reported fits.

A matched optimization study retained L-BFGS as the production outer optimizer. Pure projected Adam at two learning rates failed the joint outer-gradient and posterior-mode convergence criteria on nearly every tested fit and changed some LRT statistics by hundreds; an Adam warm-up followed by L-BFGS recovered convergence but reached different local optima and changed statistics by as much as 3.79. Muon is not used because the fixed-effect vector has no natural matrix axis, its apparent matrix shape would couple unrelated mice or depend on the arbitrary block-contrast basis. Momentum SGD, RMSProp, Lion, and related fixed-schedule first-order methods share Adam's batching advantage but not L-BFGS's line search or curvature history, so the failed Adam experiment provides no reason to replace the converged optimizer with them. Projecting a previously fitted alternative into later null models reduced iterations but also moved local optima. Running two independent L-BFGS workers reduced matched wall time by 38% for mouse and 24% for human, with greater peak memory and little reduction in aggregate CPU time.

Genome-wide shape census supports batched likelihood evaluation but not one globally padded tensor. Among 4,825 mouse and 11,597 human block tests, only 813 and 434 belong to an exact tensor-shape group of size at least two. Allowing pseudobulk and mouse counts to vary while holding the isoform, EC, and fixed-effect dimensions constant raises these counts to 3,726 and

7,971. Cost-sorted padded batches incur estimated likelihood-work inflation of 1.97-fold for mouse and 2.11-fold for human at batch size eight, increasing with batch size. On representative low-padding batches, a padded likelihood-gradient kernel was 2.5-fold faster than a `ragged_dot` kernel for mouse and 2.2-fold faster for human. On deliberately high-padding tails, the ragged kernel was 1.09-fold and 1.86-fold faster and compiled faster.

A full padded Laplace prototype gives every fit an independent L-BFGS history, projected gradient, active state, and Armijo line search while evaluating the separable objectives together. It preserves implicit differentiation of each posterior mode and reuse of a shared unrestricted alternative. In ordinary low-dimensional buckets, batches of eight were 5.48-fold faster for mouse and 6.43-fold faster for human than sequential scalar L-BFGS-B. All eight batched nested fits converged in each dataset, compared with seven mouse and eight human scalar fits. Among fits converged by both implementations, the maximum absolute LRT-statistic differences were 0.0090 and 0.00058. A four-fit high-dimensional mouse bucket was 1.53-fold faster after alternative reuse and gave identical statistics. A high-dimensional human bucket with ten isoforms, 18 null coefficients, 252 alternative coefficients, and roughly 900 pseudobulks was 0.62-fold as fast, meaning batching was 1.62-fold slower; the two commonly converged tests differed by 6.88 and 14.80 statistic units because the optimizers reached different local optima. Active masks preserve accepted fits but do not remove the arithmetic of inactive padded lanes from a compiled vectorized evaluation.

The initial selective policy was tested on every cell-type block, using cost-sorted batches of eight only for padding ratios at most two and nested models with at most 128 outer coefficients. The mouse hybrid reduced elapsed worker time only from 31.18 to 30.48 hours, while the human hybrid increased it from 98.39 to 125.21 hours. Peak worker memory fell from 3.97 to 1.40 GiB in mouse and from 6.55 to 4.19 GiB in human. The memory reduction did not compensate for the human slowdown or the movement between local optima. Production scalar fitting preserves bounded compilation reuse within a gene and covered pseudobulk set and initializes the first unrestricted alternative from its fitted null. The initial global shape-and-cost scheduler instead broke gene-local compilation reuse and fit deduplicated alternatives from their default initialization. The latter difference moved optima even on routes where both fits were scalar, while batching introduced additional trajectories. That initial hybrid was rejected without structured-null calibration because it failed the prior runtime and numerical-agreement gates.

A follow-up restored the production null-to-alternative initialization and made counts, EC maps, fixed-effect tensors, and cluster layouts dynamic arguments to a bounded global shape cache. On 20 real mouse tests, global caching reduced elapsed time by 10.3% and objective-evaluation time by 9.7% with identical statistics, at the cost of 11.0% greater peak memory, so this bounded cache remains available. Reusing each accepted L-BFGS iterate's posterior mode with adaptive Newton termination halved optimizer time, but maximum statistic drift remained 0.319 at mode tolerance 10^{-5} and 0.187 at 10^{-8}. A plug-in raw-EC efficient-score screen was slower than the full nested LRT because forming the observed outer Hessian dominated its cost; seven of 20 fixed-effect Hessians were indefinite, and one extreme score disagreed with the LRT tail. The production-initialized GPU scheduler later reduced mouse worker time by 21.3% but increased human worker time by 36.9%, increased peak memory in both datasets, and retained local-optimum drift. Adaptive mode reuse, the score screen, and batched schedulers were therefore removed rather than exposed as analysis options.

The multinomial EC likelihood permits a cheaper exact derivative kernel than nested automatic differentiation. For one pseudobulk and primer, let $\boldsymbol{y} \in \mathbb{N}^E$ contain counts for E equivalence classes, let $M \in \{0,1\}^{E\times K}$ map those classes to K isoforms, let $\boldsymbol{a} = \mathrm{softmax}([\boldsymbol{\eta}^T, 0]^T) \in \mathbb{R}^K$ be isoform abundance for free logits $\boldsymbol{\eta} \in \mathbb{R}^{K-1}$, let $B = M\,\mathrm{diag}(\boldsymbol{a}) \in \mathbb{R}^{E\times K}$, let $\boldsymbol{m} = B\mathbf{1}_K \in \mathbb{R}^E$, let $D = \mathbf{1}_E^T\boldsymbol{m}$, let $N = \mathbf{1}_E^T\boldsymbol{y}$, and let $\boldsymbol{c} = B^T\mathbf{1}_E \in \mathbb{R}^K$. Ignoring the multinomial coefficient, the log likelihood is

$$\ell = \boldsymbol{y}^T \log \boldsymbol{m} - N \log D.$$

Its full-logit score $\boldsymbol{g} \in \mathbb{R}^K$ and Hessian $H \in \mathbb{R}^{K \times K}$ are

$$\boldsymbol{g} = B^T(\boldsymbol{y} \oslash \boldsymbol{m}) - N\boldsymbol{c}/D, \qquad H = \operatorname{diag}(\boldsymbol{g}) - B^T \operatorname{diag}(\boldsymbol{y} \oslash \boldsymbol{m}^2)B + N\boldsymbol{c}\boldsymbol{c}^T/D^2.$$

Deleting the reference isoform's last row and column gives derivatives with respect to $\boldsymbol{\eta}$, and derivatives sum over primers. Substituting these expressions for the row-level autodiff score and Hessian leaves the Laplace objective, posterior-mode solve, implicit mode gradient, bounds, and L-BFGS convergence criteria unchanged while avoiding differentiation through an autodiff-generated Hessian.

On matched subsets of 12 mouse and 10 human tests, the analytic kernel reduced end-to-end time by factors of 1.49 and 1.48 and objective-evaluation time by factors of 1.24 and 1.48. Joint convergence was unchanged at 11 and 9 tests. Median absolute LRT differences were 2.4×10^{-11} and 2.3×10^{-13}, and maxima were 0.040 and 0.124. Peak memory fell from 2.35 to 1.33 GiB in mouse and from 4.81 to 4.32 GiB in human.

Table 6 gives the complete scalar rerun. Analytic derivatives and coarser 96-way sharding reduced repeated compilation, lowering aggregate worker time, including merging, by factors of 2.48 in mouse and 1.78 in human. Mouse joint convergence was identical and human gained one converged test. Discovery Jaccard indices were 0.997 and 0.996; all 322 analytic mouse calls were among the 323 prior calls, while human shared 1,458 calls, gained five, and lost one. Median LRT drift remained near floating-point precision, although one mouse and one human local-optimum difference reached 6.83 and 26.68. Full-run peak memory was unchanged in mouse but rose from 6.55 to 8.25 GiB in human. The analytic derivative kernel with gene-grouped L-BFGS is therefore the production scalar implementation; the biological conclusions are unchanged.

Table 6: Full-data closed-form multinomial derivative audit. A and D denote analytic and nested-autodiff row derivatives with otherwise matched scalar Laplace LRT fitting. Time is total worker hours and memory is peak GiB.

Dataset	Tests	Conv. A/D	BH A/D	Shared	Jaccard	Median $\lvert\Delta\text{LRT}\rvert$	Max $\lvert\Delta\text{LRT}\rvert$	Time A/D	Memory A/D
Mouse	4,825	4,546/4,546	322/323	322	0.997	3.64×10^{-12}	6.83	12.57/31.18	4.00/3.97
Human	11,597	10,982/10,981	1,463/1,459	1,458	0.996	1.40×10^{-9}	26.68	55.17/98.39	8.25/6.55

A bounded trust-region Newton–CG alternative used exact Hessian–vector products through the same Laplace objective. On one representative mouse test, 20 trust iterations took 90.4 seconds, including 67.3 seconds in 261 Hessian–vector products, and neither nested fit converged; L-BFGS converged in 40.5 seconds. The corresponding human comparison took 151.4 versus 54.3 seconds, with 125.4 seconds in 251 Hessian–vector products and neither trust fit converged. Larger matched trust runs were still unfinished after more than three times the analytic L-BFGS benchmark duration. Exact second-order outer optimization is therefore rejected because differentiating the implicit mode and log-determinant terms is more expensive than the saved iterations.

The production-initialized genome-wide scheduler uses batches of 32, requires at least four equal-parameter fits, admits padding ratios at most three and at most 128 outer coefficients, and sends every other shape to the same scalar Laplace LRT. Failed batched fits continue from their batched parameters and, if necessary, restart from the production initialization. Table 7 gives the complete rerun. Mouse worker time fell 21.3%, from 31.18 to 24.55 hours, while peak memory rose from 3.97 to 5.31 GiB. Human worker time rose 36.9%, from 98.39 to 134.74 hours, and peak memory rose from 6.55 to 8.14 GiB. Only 1,344 of 11,597 human nulls met the batching gate; its 10,253 scalar nulls consumed 85.23 hours, so faster unrestricted alternatives could not recover the null-tail cost. The mouse and human discovery-set Jaccard indices against harmonized scalar production were 0.973 and 0.991. Median absolute LRT drift was small, 0.000131 and 0.000191, but maximum differences of 423.7 and 411.6 show that independently converged optimizers occasionally reached different local optima. A common production replacement is therefore rejected, and the reported results retain gene-grouped scalar fitting in both datasets.

Four multilevel mouse null families independently permuted the complete cell-type label vector within each subject while holding pseudobulk EC counts, nuisance columns, covered rows, and within-subject label counts fixed. Among 18,379 converged hybrid tests, rejection rates were 0.0983, 0.0385, and 0.0141 at nominal levels 0.05, 0.01, and 0.001, and every family produced 51–64 BH calls. An exact gene-grouped scalar rerun of the first permutation gave rates 0.1047, 0.0432, and 0.0148 and 62 BH calls, compared with 64 hybrid calls, 60 shared, and discovery Jaccard 0.909. Thus batching does not cause the null inflation, but the ordinary multilevel omnibus chi-square LRT is not calibrated and its BH counts in the matched method comparison must be treated as descriptive discovery yield rather than controlled FDR. This finding does not apply to the separately bootstrap-calibrated primary mouse analysis or the paired-label pairwise benchmark.

Table 7: Production-initialized full-data selective batched L-BFGS audit. H and S denote hybrid and scalar production fitting. Drift is evaluated among tests for which both procedures converged under both nested models. Time is elapsed worker hours and memory is peak GiB.

Dataset	Tests	Conv. H/S	BH H/S	Shared	Jaccard	Median $\lvert\Delta\text{LRT}\rvert$	Max $\lvert\Delta\text{LRT}\rvert$	Time H/S	Memory H/S
Mouse	4,825	4,597/4,546	324/323	319	0.973	0.000131	423.7	24.55/31.18	5.31/3.97
Human	11,597	11,004/10,981	1,460/1,459	1,453	0.991	0.000191	411.6	134.74/98.39	8.14/6.55

Table 8: Analysis resource use for the matched comparator results. Time is aggregate worker hours and memory is peak GiB for one worker. Tealeaf totals include calibration or continuation retries used by the reported analyses.

	Mouse		Human	
Method	Time	Memory	Time	Memory
Tealeaf EC GLMM	118.8	4.74	729.4	6.83
LeafCutter	11.7	1.28	38.6	3.26
MAJIQ Heterogen	6.6	1.38	9.5	1.92
scQuint	15.3	3.99	94.1	4.76

Raw yield still does not establish that one method is better. We therefore use an independent-subject reproducibility benchmark on a common gene and cell-type-pair universe. For this benchmark Tealeaf uses a paired junction log-ratio test. It deliberately uses the same filtered junction groups as scQuint, so its advantage cannot be attributed to EC ambiguity resolution or a larger testing universe.

Paired junction log-ratio test

For one junction group, let $K \geq 2$ be its number of retained introns and let n be the number of subjects represented exactly once in each of two cell types. For subject $i \in \{1, \ldots, n\}$, cell type $c \in \{0, 1\}$, and intron $k \in \{1, \ldots, K\}$, let $y_{ick} \in \mathbb{N}_0$ be the pseudobulk junction count. Define $\boldsymbol{y}_{ic} = (y_{ic1}, \ldots, y_{icK})' \in \mathbb{N}_0^K$, the centered log-ratio

$$\operatorname{clr}(\boldsymbol{y}_{ic}) = \log(\boldsymbol{y}_{ic} + 1/2) - \frac{\mathbf{1}_K}{K} \mathbf{1}_K' \log(\boldsymbol{y}_{ic} + 1/2) \in \mathbb{R}^K,$$

and the within-subject difference $\boldsymbol{d}_i = H_K'\{\mathrm{clr}(\boldsymbol{y}_{i1}) - \mathrm{clr}(\boldsymbol{y}_{i0})\} \in \mathbb{R}^{K-1}$. Here $H_K \in \mathbb{R}^{K\times(K-1)}$ is an orthonormal Helmert basis for the subspace orthogonal to $\mathbf{1}_K$; the result is invariant to the particular orthonormal basis. Pairing removes every additive subject effect shared by the two cell types, while the log-ratio removes junction-group depth.

The null is $H_0 : \mathbb{E}(\boldsymbol{d}_i) = \mathbf{0}_{K-1}$. For $K = 2$, Tealeaf uses the exact two-sided one-sample Student t test. For $K > 2$, with $q = K - 1$, sample mean $\bar{\boldsymbol{d}} \in \mathbb{R}^q$, and sample covariance $S_d \in \mathbb{R}^{q\times q}$, it computes

$$T^2 = n\bar{\boldsymbol{d}}' S_d^{-1}\bar{\boldsymbol{d}}, \qquad \frac{n-q}{q(n-1)}T^2 \sim F_{q,n-q} \quad \text{under } H_0.$$

The test requires $n \geq 8$, $n > q$, and full-rank S_d. The eight-subject threshold was fixed from permutation calibration before the observed split comparison. Junction groups use the scQuint filters: each intron must occur in at least three samples per arm, have global proportion at least 10^{-3}, and belong to a nonsingleton group.

Algorithm 3 Paired junction log-ratio test for one cell-type pair

Require: Count vectors $\boldsymbol{y}_{ic} \in \mathbb{N}_0^K$ and subject labels.
Keep subjects with exactly one observation from each cell type and nonzero junction-group depth in both observations.
Apply the junction filters and require $n \geq 8$, $n > K - 1$.
for each retained subject i **do**
  Compute $\boldsymbol{d}_i = H_K'\{\mathrm{clr}(\boldsymbol{y}_{i1}) - \mathrm{clr}(\boldsymbol{y}_{i0})\}$.
end for
if $K = 2$ **then**
  Return the two-sided one-sample t-test p-value.
else
  Estimate S_d; mark the feature ineligible if S_d is rank deficient.
  Return the exact Hotelling T^2-to-F p-value.
end if

Independent-subject power comparison

The benchmark divides biological subjects into two condition-balanced halves and reruns every cell-type analysis independently. Features are mapped to genes and feature p-values are combined within gene by Simes. Each Tealeaf–comparator evaluation is restricted to genes and cell-type pairs eligible for both methods in both halves. For method m, gene g, and fold $h \in \{1, 2\}$, let $p_{mgh} \in [0, 1]$ be the Simes gene p-value. The conjunction p-value

$$p_{mg}^{\mathrm{rep}} = \max(p_{mg1}, p_{mg2})$$

tests whether the gene has signal in both independent halves. BH at 0.05 is applied once to these conjunction p-values on the shared gene universe. Secondary endpoints are the fraction of nominal discoveries in one half with $p \leq 0.05$ in the other and Spearman correlation of $-\log_{10} p$ across all shared gene–pair tests.

Table 9 gives the uncalibrated comparator results. Tealeaf finds more independently replicated genes than every comparator: 5.4 times as many as LeafCutter, 4.2 times as many as scQuint, and 1.8 times as many as MAJIQ. The MAJIQ universe is small, so its comparison is less precise. Tealeaf also has higher held-half replication than LeafCutter and MAJIQ and markedly higher fold-to-fold rank correlation than LeafCutter and scQuint.

Table 9: Cell-type differential-splicing reproducibility across independent subject halves. "Rep." is the number of genes passing BH on p^{rep}; "held" is held-half nominal replication; ρ is Spearman correlation of gene–pair $-\log_{10} p$. Comparator p-values in this table are their native, uncalibrated values.

	Shared	Tealeaf			Comparator		
Comparator	genes	Rep.	held	ρ	Rep.	held	ρ
LeafCutter	264	193	0.906	0.685	36	0.712	0.449
MAJIQ Heterogen	12	11	0.958	0.713	6	0.875	0.769
MAJIQ relaxed UMI	1,630	490	0.791	0.638	50	0.723	0.369
scQuint	2,164	646	0.788	0.642	154	0.819	0.435

Calibration uses 32 paired-label-swap families. Within each cell-type pair, subjects are swapped independently, conditional on at least one swapped and one unswapped subject; this excludes the identity and complete-reversal configurations, both of which reproduce the observed two-sided contrast. Across 12,760 eligible Tealeaf null tests, the rejection rates at $0.05, 0.01, 0.001, 10^{-4}$ are respectively $0.0509, 0.00964, 0.000627, 0.0000784$. Reapplying BH within each permuted family yields two discoveries across 32 families. Thus the native Tealeaf p-values are adequately calibrated at the selected threshold.

The comparator nulls are not uniformly calibrated: their pooled nominal 0.05 rates are 0.0620 for LeafCutter, 0.0837 for scQuint, and 0.0260 for MAJIQ. As a sensitivity analysis, each comparator p-value is mapped through its method-specific pooled empirical null CDF before repeating the same benchmark. Tealeaf remains unchanged and has 193 versus 21 replicated genes for LeafCutter, 646 versus 134 for scQuint, and 11 versus 0 for MAJIQ. The power conclusion therefore does not depend on favorable comparator miscalibration; correcting it increases Tealeaf's margin. The 32-family calibration is intended to validate the body and moderate tail of each null, not to estimate genome-wide p-values smaller than its empirical resolution.

The stringent MAJIQ universe is small because its combined build and quantification defaults leave only 46 LSV edges from 13 genes; the cross-fold, common-pair intersection contains 12 genes. Lowering only the Heterogen experiment fraction increases feature–contrast tests but cannot add features absent from PSI coverage. Rebuilding with the relaxed UMI thresholds produces 20,954 and 18,914 fold-specific LSV features and a common comparison of 1,630 genes. Its 178,793 null edge tests reject at rates 0.0320, 0.00456, and 0.000151 at nominal levels 0.05, 0.01, and 0.001, with no within-family BH discoveries in 32 permutations. Empirical-null calibration corrects this conservatism. On the expanded universe, Tealeaf has 490 replicated genes versus 50 for native MAJIQ and 89 for calibrated MAJIQ. The power advantage therefore persists after removing the stringent MAJIQ coverage bottleneck.

A Alternative Models and Supporting Derivations

The main text defines the two primary EC-count GLMMs and their implemented test. This appendix records the path-estimate models, covariance approximations, alternative EC likelihoods, and inference benchmarks used to understand or validate that analysis.

Paired-primer EC likelihood

For block b, pseudobulk i, and primer p, define $N_{bip} = \mathbf{1}'_{E_{bp}} \boldsymbol{c}_{bip} \in \mathbb{N}_0$, $\boldsymbol{q}_{bip} = A_{bp}\boldsymbol{\theta}_{bi} \in \mathbb{R}_+^{E_{bp}}$, and $Z_{bip} = \mathbf{1}'_{E_{bp}} \boldsymbol{q}_{bip} \in \mathbb{R}_+$. Conditional on the primer-specific UMI total,

$$\boldsymbol{c}_{bip} \mid N_{bip}, \boldsymbol{\theta}_{bi} \sim \operatorname{Multinomial}\left(N_{bip}, \boldsymbol{q}_{bip}/Z_{bip}\right).$$

The oligo(dT) and random-hexamer protocols use separate A_{bp} matrices and likelihood terms. Their different isoform exposures therefore model primer-specific positional coverage without forcing a common bias profile.

Pre-collapsed local-path comparator

The path-space comparator reduces the latent dimension before fitting but retains the raw EC count vectors. Let $T_g \in \mathbb{N}$ be the supported isoform count, $S_b \in \mathbb{N}$ the represented local-path count, and $U_b \in \mathbb{N}_0$ the number of isoforms retained only as normalization nuisances. It first estimates one label-independent pooled isoform mixture $\widehat{\boldsymbol{w}}_g \in \Delta^{T_g-1}$ by maximizing the conditional EC log likelihood,

$$\widehat{\boldsymbol{w}}_g = \arg\max_{\boldsymbol{w}\in\Delta^{T_g-1}} \sum_p \left\{ \sum_{e=1}^{E_{gp}} \bar{c}_{gpe} \log(A_{gp}\boldsymbol{w})_e - \bar{N}_{gp} \log(\mathbf{1}'_{E_{gp}} A_{gp}\boldsymbol{w}) \right\},$$

where $\bar{\boldsymbol{c}}_{gp} = \sum_i \boldsymbol{c}_{gip} \in \mathbb{N}_0^{E_{gp}}$ and $\bar{N}_{gp} = \mathbf{1}'_{E_{gp}} \bar{\boldsymbol{c}}_{gp} \in \mathbb{N}_0$. This pooling uses no cell-type or condition labels.

For represented path $s \in \{1, \ldots, S_b\}$, let $\mathcal{I}_{bs} \subseteq \{1, \ldots, T_g\}$ be its isoforms and define the fixed conditional mixture $\widehat{\rho}_{bst} = \widehat{w}_{gt} / \sum_{u\in\mathcal{I}_{bs}} \widehat{w}_{gu}$ for $t \in \mathcal{I}_{bs}$. The collapsed primer map $B_{bp} \in \mathbb{R}_+^{E_{gp}\times(S_b+U_b)}$ has represented-path columns

$$(B_{bp})_{\cdot s} = \sum_{t\in\mathcal{I}_{bs}} \widehat{\rho}_{bst} (A_{gp})_{\cdot t} \in \mathbb{R}_+^{E_{gp}},$$

while every nuisance isoform retains its original column separately. If $\boldsymbol{\phi}_{bi} \in \mathbb{R}_+^{S_b+U_b}$ is the reduced latent abundance, the comparator replaces $A_{gp}\boldsymbol{\theta}_{gi}$ by $B_{bp}\boldsymbol{\phi}_{bi}$ and otherwise uses the same multinomial EC likelihood, subject random intercept, fixed-effect construction, Laplace fitting procedure, likelihood-ratio reference distribution, and BH correction as the isoform-space model.

Because counts are not assigned to paths before fitting, this construction retains the multinomial EC covariance $N_{bip}\{\operatorname{diag}(\boldsymbol{\pi}_{bip}) - \boldsymbol{\pi}_{bip}\boldsymbol{\pi}'_{bip}\} \in \mathbb{S}_+^{E_{gp}}$ and propagates EC ambiguity through B_{bp}. It does not retain isoform-specific information within a path. The reduction is exact only when the conditional isoform mixture is constant across pseudobulks and the fixed mixture is correct; otherwise it is a lower-dimensional working model. Nuisance isoforms remain separate specifically to avoid imposing one mixture on heterogeneous gene-level isoforms outside the tested block.

This reduction is small in the fitted universes. Only 29.7% of mouse block tests and 12.8% of human block tests contain two represented isoforms that can be collapsed to the same path. The mean fractional latent-dimension reductions are 10.6% and 4.4%, respectively. Among multi-isoform mouse path-by-primer groups in the two subject folds, only 0.22% and 0.24% have identical compatibility columns, confirming that the reduction is generally a working-model approximation rather than an exact reparameterization.

Table 10 compares the two latent spaces using the same nested-autodiff Laplace implementation available when the ablation was run. Discovery is reported both on each method's converged tests and on the common-converged universe. Independent-subject replication requires a block test to converge for both methods in both folds, combines block p-values to genes by Simes, and applies BH to the two-fold conjunction p-value. The human sensitivity analysis uses 359 matched completed shards containing 4,282 tests and is not a genome-wide human analysis. Path space produces more discovery calls in both datasets but only two more replicated mouse genes, slightly higher held-half replication, and lower fold rank correlation. It is also slower and uses slightly more peak memory. Thus pre-collapse does not provide a clear power or computational advantage. The closed-form derivative audit was not repeated for path space; its isoform-space rerun changes the genome-wide mouse count by one call and supersedes the isoform-space resource entries, but does not change this ablation's conclusion.

Table 10: Isoform-space and pre-collapsed path-space EC GLMM comparison. Discovery rates use BH calls divided by eligible tests. Rep. is the gene count passing BH on the two-fold conjunction p-value, held is held-half nominal replication, and ρ is fold Spearman correlation. Time is aggregate worker hours and memory is peak GiB.

Endpoint	Isoform space	Path space
Mouse convergence	0.942	0.949
Mouse BH calls, method-specific	323	354
Mouse discovery rate, method-specific	0.0711	0.0773
Mouse BH calls, common converged	319	348
Mouse discovery rate, common converged	0.0710	0.0774
Replicated genes	37	39
Held-half replication	0.676	0.682
Fold rank correlation ρ	0.433	0.395
Mouse discovery worker hours	31.18	33.23
Mouse discovery peak GiB	3.97	4.04
Human subset convergence	0.949	0.949
Human subset BH calls	477	506
Human subset discovery rate	0.1174	0.1246
Human subset worker hours	33.69	36.55
Human subset peak GiB	6.39	6.65
Fold 0 worker hours	8.17	8.40
Fold 1 worker hours	7.60	8.26

Estimate-once path Wald ablation

This subsection records a rejected computational experiment. Its analytic reference failed the structured-null audit, so the estimate-once Wald implementation and its analysis launchers were removed; the result tables and findings remain for provenance.

The estimate-once ablation removes repeated raw-EC null and alternative GLMM fits. For each block b, it uses the same pooled, label-independent isoform baseline $\widehat{\boldsymbol{w}}_g \in \Delta^{T_g-1}$ defined above. For pseudobulk $i \in \{1,\ldots,n_b\}$, it estimates one path perturbation $\widehat{\boldsymbol{\delta}}_{bi} \in \mathbb{R}^{d_b}$, where $d_b = S_b - 1 \in \mathbb{N}$, by maximizing the two-primer conditional EC likelihood while holding the within-path isoform mixture fixed. The resulting path composition is $\widehat{\boldsymbol{\psi}}_{bi} \in \Delta^{S_b-1}$, its ILR response is $\widehat{\boldsymbol{y}}_{bi} = Q_b' \log \widehat{\boldsymbol{\psi}}_{bi} \in \mathbb{R}^{d_b}$, and the inverse conditional Fisher information gives $V_{bi} \in \mathbb{S}_+^{d_b}$. Thus the full off-diagonal covariance among path log ratios induced by the multinomial EC likelihood is retained; ECs are not assigned independently to paths.

The second stage uses the same nuisance and tested fixed-effect columns as the EC GLMM. Let $X_b \in \mathbb{R}^{n_b\times k_b}$ contain both sets of columns and $m(i) \in \{1,\ldots,M_b\}$ map pseudobulk i to one of $M_b \in \mathbb{N}$ mice. Define $D_{bi} = X_{b,i\cdot} \otimes I_{d_b} \in \mathbb{R}^{d_b\times k_b d_b}$, $W_{bi} = V_{bi}^{-1} \in \mathbb{S}_+^{d_b}$, and $\boldsymbol{\beta}_b = \mathrm{vec}(\boldsymbol{B}_b) \in \mathbb{R}^{k_b d_b}$, where $\boldsymbol{B}_b \in \mathbb{R}^{k_b\times d_b}$. The precision-weighted estimate is $\widehat{\boldsymbol{\beta}}_b = A_b^{-1}\sum_i D_{bi}' W_{bi} \widehat{\boldsymbol{y}}_{bi}$, where $A_b = \sum_i D_{bi}' W_{bi} D_{bi} \in \mathbb{S}_+^{k_b d_b}$. A pseudobulk whose EC information does not identify every tested path coordinate receives zero participation; a block is ineligible when the remaining rows do not span X_b or $M_b \le k_b$.

Mouse dependence and biological variation enter through a clustered sandwich covariance rather than a fitted latent Gaussian. For mouse m, define its score $\boldsymbol{s}_{bm} = \sum_{i:m(i)=m} D'_{bi} W_{bi}(\widehat{\boldsymbol{y}}_{bi} - D_{bi}\widehat{\boldsymbol{\beta}}_b) \in \mathbb{R}^{k_b d_b}$. The coefficient covariance is $\widehat{\Sigma}_b = c_b A_b^{-1}(\sum_m \boldsymbol{s}_{bm}\boldsymbol{s}'_{bm})A_b^{-1} \in \mathbb{S}_+^{k_b d_b}$, where $c_b \in \mathbb{R}_+$ is the standard cluster and residual degrees-of-freedom correction. If the tested covariance has rank $q_b \in \mathbb{N}$, the reference distribution is $W_b/q_b \sim F_{q_b, M_b - k_b}$, which reduces the small-cluster tail inflation of the chi-square reference.

This is a two-stage working model, not an equivalent reparameterization of the EC GLMM. It does not propagate uncertainty in $\widehat{\boldsymbol{w}}_g$, assumes the local likelihood is adequately Gaussian on the ILR scale, freezes the observation covariance before estimating biological effects, and represents mouse dependence marginally rather than with a parametric random intercept. Its purpose is to measure the speed and empirical power available when raw EC ambiguity is resolved once rather than revisited inside every random-effect mode and optimizer iteration.

In the original nested-autodiff mouse cell-type ablation, the estimate-once test was eligible for 4,499 of 4,825 blocks and made 3,069 uncalibrated BH discoveries, a 68.2% discovery rate, compared with 323 of 4,546 for the isoform-space EC GLMM and 354 of 4,579 for the collapsed-path EC GLMM. On the 4,211 blocks eligible for all three methods, their discovery counts were 303, 335, and 2,876. Of the 303 isoform-space discoveries, 276 were also selected by estimate-once Wald, but its much larger discovery set gave a Jaccard index of 0.095. In the subject split, 948 blocks and 724 genes were testable by all three methods. Isoform-space EC GLMM, collapsed-path EC GLMM, and estimate-once Wald produced 35, 40, and 276 genes significant in both halves; their held-half nominal replication rates were 0.669, 0.685, and 0.751, and their fold rank correlations were 0.429, 0.393, and 0.426. The structured-null audit below shows that these Wald discovery and block-replication counts are inflated and must not be interpreted as evidence of additional power.

On the human cell-type analysis, 8,062 of 11,597 blocks were eligible and 5,906 passed uncalibrated BH, a 73.3% discovery rate, whereas the isoform-space EC GLMM was eligible for 11,003 blocks and selected 1,487, a 13.5% rate. On 7,685 common-eligible blocks, the methods selected 1,170 and 5,633; 873 calls overlapped and the discovery-set Jaccard index was 0.147. The human eligibility loss comes mainly from requiring the complete local-path ILR vector and covariance to be identifiable in each retained pseudobulk, while the high discovery rate reflects the same null failure seen in mouse.

Including the small correction reruns used to enforce positive denominator degrees of freedom, estimate-once Wald used 2.78 aggregate worker hours and 0.54 GiB peak memory for mouse cell type, versus 12.57 hours and 4.00 GiB for the analytic-derivative isoform-space EC GLMM. Human values were 14.18 hours and 3.58 GiB, versus 55.17 hours and 8.25 GiB. The full external-method campaigns in Table 8 used 6.61–15.28 mouse worker hours and 9.55–94.09 human worker hours. Those campaigns cover different feature units and broader reported effect families, so they are contextual rather than matched denominators; nevertheless, resolving each EC likelihood once moves runtime into the comparator range. Retaining a dense $d_b \times d_b$ covariance is therefore not the main cost. The primary EC GLMM remains slower because it repeatedly evaluates the raw EC likelihood and solves the latent Gaussian problem under both nested models.

The initial balanced two-level Gaussian null simulation did not reproduce the leverage and tested dimensions of the real design. The structured audit therefore holds every estimated $\widehat{\boldsymbol{y}}_{bi}$, V_{bi}, nuisance column, and mouse membership fixed and independently shuffles the observed cell-type labels within each mouse. It also uses a restricted wild-cluster null, fitting the nuisance-only precision-weighted model and multiplying its complete multivariate residual vector by one

Rademacher sign per mouse. Across 32 mouse null families, the CR1 analytic p-values rejected 65.8%, 58.3%, and 51.7% of label permutations at nominal 0.05, 0.01, and 0.001. Wild-cluster rates were 60.3%, 53.2%, and 46.8%. Each permuted family produced a mean 2,844 BH calls and each wild family a mean 2,581, compared with 3,069 in the observed data. Mapping observed analytic p-values through the matching null CDF within numerator-degree and cluster-degree strata left zero BH calls for either null construction.

A CR3 sensitivity premultiplies each whitened mouse score by the inverse cluster hat-matrix complement, rather than applying only the CR1 scalar correction. It reduced the observed mouse BH count from 3,069 to 1,906 but remained anti-conservative. Across 16 null families its nominal 0.05 rejection rates were 36.7% for label permutations and 33.8% for wild nulls, with mean BH counts 1,430 and 1,277. Stratified null calibration again left zero BH calls. The failure increases with the joint tested dimension, but CR3 one-degree-of-freedom tests still reject roughly 12–24% under the structured null, so restricting to smaller blocks does not solve the problem.

An evenly spaced human sensitivity used 11 of 96 shards, 912 eligible observed blocks, and 16 null families. Label-permutation rejection was 75.3%, 72.3%, and 68.7% at nominal 0.05, 0.01, and 0.001; wild-null rejection was 72.3%, 68.8%, and 65.2%. The null families averaged 666 and 652 BH calls, and matched stratified calibration left zero. Thus the failure generalizes to human data and is stronger in its higher-dimensional tests.

Effect direction provides a separate check that does not depend on the analytic reference distribution. For each shared cell-type pair and local path, the two fold-specific ILR coefficient matrices were transformed to centered-log path effects. Across 985 blocks and 14,059 matched pair-by-path coordinates, signs agreed in 56.8% and the fold effect Spearman correlation was 0.151. Coordinates in blocks passing uncalibrated BH in both folds had 57.7% agreement and correlation 0.138, versus null sign agreement 51.0% for label permutations and 50.2% for wild draws. Requiring $|z| \geq 1.96$ in both folds raised agreement to 61.6%, but only 1,035 coordinates from 279 blocks remained and their correlation was 0.173. Under CR3, the corresponding strong-coordinate agreement was 80.6% among 217 coordinates from 69 blocks, but CR3 p-values themselves remain invalid. These results retain the runtime conclusion but reject estimate-once sandwich Wald as an inferential replacement for the EC GLMM.

Gaussian mixed-model LRT and score Bayes factors

This subsection likewise records a rejected inferential experiment. Neither the LRT tail nor fixed Bayes-factor thresholds were calibrated well enough to replace the EC GLMM, so these implementations were removed from the package and analysis runners.

Two alternatives retain the estimate-once path responses but replace the clustered sandwich calculation. Stack the usable responses as $\widehat{\boldsymbol{y}}_b \in \mathbb{R}^{n_b d_b}$, let $Z_b \in \mathbb{R}^{n_b d_b \times a_b}$ be the nuisance design after taking its Kronecker product with I_{d_b}, and let $T_b \in \mathbb{R}^{n_b d_b \times q_b}$ be the corresponding tested design, where $a_b \in \mathbb{N}$ and $q_b \in \mathbb{N}$ are the nuisance and tested coefficient dimensions. The Gaussian random-intercept model is

$$\widehat{\boldsymbol{y}}_b = Z_b \boldsymbol{\alpha}_b + T_b \boldsymbol{\gamma}_b + U_b \boldsymbol{u}_b + \boldsymbol{\epsilon}_b, \qquad \boldsymbol{u}_b \sim N(\mathbf{0}, \sigma^2_{m,b} I_{M_b d_b}), \qquad \boldsymbol{\epsilon}_b \sim N\{\mathbf{0}, \mathrm{blockdiag}_i(V_{bi} + \sigma^2_{e,b} I_{d_b})\},$$

where $U_b \in \{0,1\}^{n_b d_b \times M_b d_b}$ maps responses to mouse-specific d_b-vectors, $\boldsymbol{\alpha}_b \in \mathbb{R}^{a_b}$, $\boldsymbol{\gamma}_b \in \mathbb{R}^{q_b}$, $\boldsymbol{u}_b \in \mathbb{R}^{M_b d_b}$, and $\boldsymbol{\epsilon}_b \in \mathbb{R}^{n_b d_b}$. Consequently, responses from the same mouse have cross-pseudobulk covariance $\sigma^2_{m,b} I_{d_b}$, while $\sigma^2_{e,b} \in \mathbb{R}_+$ absorbs biological variation not represented by the conditional EC covariance V_{bi}.

Let $X_{0b} = Z_b$, $X_{1b} = [Z_b \; T_b]$, and let $\Omega_b(\sigma^2_m, \sigma^2_e) \in \mathbb{S}^{n_b d_b}_{++}$ be the marginal covariance implied by the model. For model $h \in \{0,1\}$, maximum likelihood profiles its fixed effects as $\widehat{\boldsymbol{\xi}}_{hb} =$

$(X'_{hb}\Omega_b^{-1}X_{hb})^{-1}X'_{hb}\Omega_b^{-1}\widehat{\boldsymbol{y}}_b$ and separately optimizes both nonnegative variance components using the profile objective

$$-2\ell_{hb} = \log|\Omega_b| + (\widehat{\boldsymbol{y}}_b - X_{hb}\widehat{\boldsymbol{\xi}}_{hb})'\Omega_b^{-1}(\widehat{\boldsymbol{y}}_b - X_{hb}\widehat{\boldsymbol{\xi}}_{hb}) + n_b d_b \log(2\pi).$$

This is ML rather than REML because the nested models have different fixed-effect dimensions. The likelihood-ratio statistic is $\Lambda_b = 2(\ell_{1b} - \ell_{0b}) \in \mathbb{R}_+$, with candidate reference $\chi^2_{q_b}$, and the cluster-count BIC approximation is $\log \mathrm{BF}^{\mathrm{BIC}}_{10,b} = \{\Lambda_b - q_b \log M_b\}/2$. Both variance components are refitted under the alternative, including boundary candidates with either or both components equal to zero.

The implementation does not form or factor a dense covariance for all pseudobulks from one mouse. Within mouse m, let $R_{bm} = \mathrm{blockdiag}_{i:m(i)=m}(V_{bi} + \sigma^2_{e,b}I_{d_b})$ and $K_{bm} = \mathbf{1} \otimes I_{d_b}$. It applies $(R_{bm} + \sigma^2_{m,b}K_{bm}K'_{bm})^{-1}$ with the Woodbury identity and computes its determinant through the $d_b \times d_b$ matrix $I_{d_b} + \sigma^2_{m,b}K'_{bm}R^{-1}_{bm}K_{bm}$. Runtime therefore scales with individual path-covariance factors and a path-dimensional correction rather than a cubic factorization in the number of pseudobulks per mouse.

The score Bayes factor requires only the fitted null covariance $\widehat{\Omega}_{0b}$. Define the nuisance-residual precision $P_{0b} = \widehat{\Omega}^{-1}_{0b} - \widehat{\Omega}^{-1}_{0b}Z_b(Z'_b\widehat{\Omega}^{-1}_{0b}Z_b)^{-1}Z'_b\widehat{\Omega}^{-1}_{0b} \in \mathbb{S}^{n_b d_b}_{+}$, the efficient score $\boldsymbol{s}_b = T'_b P_{0b}\widehat{\boldsymbol{y}}_b \in \mathbb{R}^{q_b}$, and its information $J_b = T'_b P_{0b} T_b \in \mathbb{S}^{q_b}_{++}$. For a proper alternative prior $\boldsymbol{\gamma}_b \sim N(\mathbf{0}, W_{bs})$, the local Gaussian integral is

$$\log \mathrm{BF}_{10,b}(s) = \frac{1}{2}\left\{\boldsymbol{s}'_b(J_b + W^{-1}_{bs})^{-1}\boldsymbol{s}_b - \log|I_{q_b} + W_{bs}J_b|\right\}.$$

The first term rewards evidence away from the null and the determinant is the prior-volume penalty.

The production sensitivity averages equal prior mass over $s \in \{0.1, 0.25, 0.5, 1.0\}$, measured as the standard deviation of one latent cell-type effect in path-ILR units. If the tested cell-type design contains $L_b - 1$ reference contrasts, independent latent effects with variance s^2 induce the reference-invariant covariance $W_{bs} = s^2\{(I_{L_b-1} + \mathbf{11}') \otimes I_{d_b}\} \in \mathbb{S}^{q_b}_{++}$. The reported mixture is $\log \mathrm{BF}^{\mathrm{mix}}_{10,b} = \mathrm{logsumexp}_s\{\log \mathrm{BF}_{10,b}(s)\} - \log 4$. This avoids changing the evidence when a different reference cell type is chosen. It does not repair misspecification of the estimated-path Gaussian model, so both the LRT p-values and Bayes-factor thresholds are subjected to the same within-mouse label permutations and restricted wild-cluster nulls as the Wald ablation.

On the full mouse analysis, the Gaussian path model was eligible for 4,472 of 4,825 blocks and made 200 analytic BH discoveries, 4.47% of eligible blocks. The score-mixture Bayes factor exceeded 10 for 277 blocks and 100 for 163, while the BIC Bayes factor exceeded those thresholds for 83 and 51. On 4,232 blocks also eligible for the isoform-space EC GLMM, the EC model selected 305 and the Gaussian path LRT selected 195, with 105 shared calls and discovery-set Jaccard index 0.266. This is much less discrepant than the estimate-once sandwich Wald analysis, but nominal thresholds still require the structured-null check below.

All 96 mouse shards completed the 32 within-mouse label permutations. Their pooled LRT rejection rates were 4.16%, 1.51%, and 0.478% at nominal 0.05, 0.01, and 0.001, and one null family produced a mean 11.0 analytic BH calls. Thus the center is conservative but the deep tail is inflated. The restricted wild-cluster sensitivity covered 92 shards and gave pooled rates 2.24%, 0.638%, and 0.150%, with a mean 1.56 BH calls. Directly estimating the expected false-discovery count from each complete label-permutation family retained 12 LRT calls at empirical FDR 0.05, compared with 194 under the wild null; the conservative intersection therefore contains 12. The corresponding counts were 12 and 269 for the score-mixture Bayes factor and 27 and 99 for the BIC approximation. At the fixed threshold $\mathrm{BF} \geq 10$, label permutations predict 28.3 null score-mixture calls among 4,472 tests versus 277 observed, and 7.94 null BIC calls versus 83 observed.

Bayes factors therefore rank a small supported tail, especially after the BIC dimension penalty, but neither the score approximation nor the LRT supplies calibrated analytic thresholds. The production EC GLMM remains the primary inferential model; estimate-once Gaussian results are an ablation or must use design-matched empirical calibration.

Across 963 mouse block tests and 725 genes shared between both subject halves and the EC GLMM, the isoform-space model had 35 genes passing BH in both halves, held-half nominal replication 0.668, and fold log-p correlation 0.432. The Gaussian path LRT had 17 replicated genes, held-half replication 0.724, and fold correlation 0.716. Among 14,014 matched cell-type-pair by path coordinates, overall sign agreement was 0.573 and effect correlation was 0.148. Requiring $|z| \geq 1.96$ in both halves retained 171 coordinates from 54 blocks, with sign agreement 0.947 and correlation 0.932. The strong effects are more directionally stable than those selected by the sandwich Wald statistic, although this does not establish tail calibration.

The dense mouse run used 4.46 aggregate worker hours. Woodbury used 4.83 hours and therefore did not improve aggregate mouse time, although it reduced the slowest shard from 502 to 418 seconds. On an identical 20-block high-dimensional human estimate set, Woodbury reduced runtime from 193.6 to 168.0 seconds, 13.2%, and changed the largest LRT statistic by only 2.2×10^{-7}. The Woodbury implementation is retained because its memory and factorization dimension do not grow cubically with the number of pseudobulks per mouse.

An evenly spaced human sensitivity used 11 of 96 shards, with 935 eligible blocks among 1,329 candidates. The analytic LRT made 153 BH calls, the score-mixture Bayes factor exceeded 10 for 196 blocks, and the BIC Bayes factor exceeded 10 for 19. Across eight within-subject label-permutation families, LRT rejection was 6.08%, 3.37%, and 2.09% at nominal 0.05, 0.01, and 0.001. The first threshold is near nominal but the extreme tail remains inflated. Empirical tail-FDR estimates retained 37 LRT, 32 score-mixture, and 19 BIC-evidence calls at 0.05, but eight null families give a coarse sensitivity rather than a definitive genome-wide calibration.

The genome-wide hierarchical model supplies $\boldsymbol{\theta}_{bi}^{(0)} \in \mathbb{R}_{+}^{T_g}$. Let $s_b(t) \in \{1, \dots, S_b\}$ map isoform $t \in \mathcal{I}_b^{\mathrm{sp}}$ to its path. The baseline spliced-isoform mass is the scalar $\omega_{bi}^{\mathrm{sp}} = \sum_{t \in \mathcal{I}_b^{\mathrm{sp}}} \theta_{bit}^{(0)} \in \mathbb{R}_{+}$. For $t \in \mathcal{I}_b^{\mathrm{sp}}$, local refinement uses

$$\theta_{bit}(\boldsymbol{\delta}_{bi}) = \omega_{bi}^{\mathrm{sp}} \frac{\theta_{bit}^{(0)} \exp\{(Q_b)_{s_b(t),\cdot} \boldsymbol{\delta}_{bi}\}}{\sum_{u \in \mathcal{I}_b^{\mathrm{sp}}} \theta_{biu}^{(0)} \exp\{(Q_b)_{s_b(u),\cdot} \boldsymbol{\delta}_{bi}\}},$$

where $(Q_b)_{s,\cdot} \in \mathbb{R}^{1 \times d_b}$. Nuisance-isoform abundances are unchanged. This preserves total spliced mass and the relative isoform mixture within each path while estimating only $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$.

Path response and inferential covariance

Each column of $C_b \in \{0,1\}^{S_b \times T_g}$ corresponding to $t \in \mathcal{I}_b^{\mathrm{sp}}$ contains one 1 in row $s_b(t)$; nuisance isoform columns are zero. Define

$$\boldsymbol{a}_{bi} = C_b \boldsymbol{\theta}_{bi} \in \mathbb{R}_{+}^{S_b}, \qquad \boldsymbol{\psi}_{bi} = \frac{\boldsymbol{a}_{bi}}{\mathbf{1}_{S_b}' \boldsymbol{a}_{bi}} \in \Delta^{S_b - 1}, \qquad \boldsymbol{y}_{bi} = Q_b' \log \boldsymbol{\psi}_{bi} \in \mathbb{R}^{d_b}.$$

The perturbation parameterization implies $\boldsymbol{y}_{bi} = \boldsymbol{y}_{bi}^{(0)} + \boldsymbol{\delta}_{bi}$, where $\boldsymbol{y}_{bi}^{(0)} \in \mathbb{R}^{d_b}$ is computed from the baseline.

Let $I_{\delta,bi} \in \mathbb{S}_{+}^{d_b}$ be the expected Fisher information of the summed paired-primer log likelihood with respect to $\boldsymbol{\delta}_{bi}$. The conditional covariance is $V_{bi}^{\mathrm{cond}} = I_{\delta,bi}^{-1} \in \mathbb{S}_{+}^{d_b}$ when the information is positive definite. It conditions on the fitted total spliced mass, the isoform mixture within each path, and every nuisance-isoform abundance. Its only free directions are the d_b path perturbations.

Full-isoform nuisance (profile) covariance

The profile calculation is a covariance sensitivity analysis, not a second estimate of path usage. It retains the same $\widehat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}_+^{T_g}$ obtained from the constrained local fit, but asks how uncertain its path ILRs would be if all T_g isoform log-abundances were locally free. Equivalently, after a smooth reparameterization into path ILRs and $T_g - d_b$ nuisance coordinates, it approximates the path-ILR block of the inverse Fisher information after profiling the nuisance coordinates. No unconstrained isoform likelihood is optimized, so *profile* below means this local expected-Fisher approximation rather than an exact profile-likelihood fit.

Set $\boldsymbol{\eta}_{bi} = \log \widehat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}^{T_g}$, $D_{bi} = \operatorname{diag}(\widehat{\boldsymbol{\theta}}_{bi}) \in \mathbb{R}^{T_g \times T_g}$, $\boldsymbol{d}_{bp} = A'_{bp}\mathbf{1}_{E_{bp}} \in \mathbb{R}^{T_g}$, and $\widehat{\boldsymbol{q}}_{bip} = A_{bp}\widehat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}_+^{E_{bp}}$, $\widehat{Z}_{bip} = \mathbf{1}'_{E_{bp}}\widehat{\boldsymbol{q}}_{bip} \in \mathbb{R}_+$, and $\boldsymbol{r}_{0,bip} = (\boldsymbol{d}_{bp} \odot \widehat{\boldsymbol{\theta}}_{bi})/\widehat{Z}_{bip} \in \mathbb{R}^{T_g}$. The primer-specific expected information is

$$I_{\eta,bip} = N_{bip}\left[\frac{D_{bi}A'_{bp}\operatorname{diag}(\widehat{\boldsymbol{q}}_{bip}^{-1})A_{bp}D_{bi}}{\widehat{Z}_{bip}} - \boldsymbol{r}_{0,bip}\boldsymbol{r}'_{0,bip}\right] \in \mathbb{S}_+^{T_g}.$$

The paired-primer information is $I_{\eta,bi} = \sum_{p=1}^{P} I_{\eta,bip} \in \mathbb{S}_+^{T_g}$.

The covariance must be propagated through the isoform-to-path map. Let $\omega_{bi}^{\text{sp}} = \sum_{t \in \mathcal{I}_b^{\text{sp}}} \widehat{\theta}_{bit} \in \mathbb{R}_+$ and $a_{bis} = \sum_{t: s_b(t)=s} \widehat{\theta}_{bit} \in \mathbb{R}_+$. Define $R_{bi} = \partial \log \boldsymbol{\psi}_{bi}/\partial \boldsymbol{\eta}'_{bi} \in \mathbb{R}^{S_b \times T_g}$ entrywise by

$$(R_{bi})_{st} = \begin{cases} \mathbb{1}\{s_b(t) = s\}\widehat{\theta}_{bit}/a_{bis} - \widehat{\theta}_{bit}/\omega_{bi}^{\text{sp}}, & t \in \mathcal{I}_b^{\text{sp}}, \\ 0, & t \notin \mathcal{I}_b^{\text{sp}}. \end{cases}$$

Thus

$$J_{bi} = \frac{\partial \boldsymbol{y}_{bi}}{\partial \boldsymbol{\eta}'_{bi}} = Q'_b R_{bi} \in \mathbb{R}^{d_b \times T_g}.$$

The multinomial likelihood necessarily has a global-scale null direction, and EC ambiguity can create additional null directions. Write the symmetric eigendecomposition

$$I_{\eta,bi} = U_{bi}\operatorname{diag}(\boldsymbol{\lambda}_{bi})U'_{bi},$$

where $U_{bi} \in \mathbb{R}^{T_g \times T_g}$ is orthogonal and $\boldsymbol{\lambda}_{bi} \in \mathbb{R}^{T_g}$. An eigenvalue is treated as positive when $\lambda_{bit} > 10^{-8}\max_u |\lambda_{biu}|$. Let $\mathcal{P}_{bi} \subseteq \{1, \ldots, T_g\}$ index these positive eigenvalues, let $\mathcal{N}_{bi}$ index the remainder, and let $U_{bi,+} \in \mathbb{R}^{T_g \times |\mathcal{P}_{bi}|}$ and $U_{bi,0} \in \mathbb{R}^{T_g \times |\mathcal{N}_{bi}|}$ contain the corresponding eigenvectors.

The path response is identifiable only if its Jacobian annihilates every information-null direction. The implementation requires

$$\|J_{bi}U_{bi,0}\|_F \le 10^{-9}\max\{\|J_{bi}\|_F, 1\}.$$

If this fails, V_{bi}^{prof} is set to an infinite matrix and the estimate is excluded from profile testing. Otherwise,

$$V_{bi}^{\text{prof}} = J_{bi}U_{bi,+}\operatorname{diag}(\boldsymbol{\lambda}_{bi,\mathcal{P}}^{-1})U'_{bi,+}J'_{bi} = J_{bi}I_{\eta,bi}^{+}J'_{bi} \in \mathbb{S}_+^{d_b}.$$

Unlike V_{bi}^{cond}, this covariance allows EC-supported changes in within-path isoform mixtures, spliced mass, and nuisance isoforms to absorb information that would otherwise identify a path contrast. It is therefore the more conservative sensitivity calculation when those nuisance directions are weakly identified.

Cell-resolved likelihood

For cell h in pseudobulk i, let $\boldsymbol{c}_{bihp} \in \mathbb{N}_0^{E_{bp}}$, $N_{bihp} = \mathbf{1}'_{E_{bp}} \boldsymbol{c}_{bihp} \in \mathbb{N}_0$, and $\boldsymbol{\theta}_{bih}^{(0)} \in \mathbb{R}_+^{T_g}$ be its EC counts, total, and baseline. Every cell shares the same pseudobulk perturbation $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$, but its isoform vector $\boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi}) \in \mathbb{R}_+^{T_g}$ is normalized using its own baseline. The objective is the scalar

$$\ell_{bi}(\boldsymbol{\delta}_{bi}) = \sum_{h=1}^{H_{bi}} \sum_{p=1}^{P} \log \mathrm{Multinomial}\left(\boldsymbol{c}_{bihp}; N_{bihp}, \frac{A_{bp} \boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})}{\mathbf{1}'_{E_{bp}} A_{bp} \boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})} \right).$$

Thus cells contribute assignment information but do not become biological replicates. Let $U_{bih} \in \mathbb{R}_+$ be the cell's total UMI count and $g_{bih} \in [0, 1]$ its decoded proportion for the block's gene. The scalar weight is $w_{bih} = U_{bih} g_{bih}$, and the reported pseudobulk composition is

$$\boldsymbol{\psi}_{bi} = \frac{\sum_{h=1}^{H_{bi}} w_{bih} \boldsymbol{\psi}_{bih}}{\sum_{h=1}^{H_{bi}} w_{bih}} \in \Delta^{S_b - 1}.$$

Cell- and primer-specific Fisher information matrices in $\mathbb{S}_+^{d_b}$ are summed, inverted, and propagated through this weighted average to obtain $V_{bi} \in \mathbb{S}_+^{d_b}$.

Effective counts and compositional regression

Treating all gene UMIs as known path assignments is overconfident when ECs are ambiguous. For each fitted $\boldsymbol{\psi}_{bi}$, define the unit-multinomial ILR covariance

$$G_{bi} = Q_b' \,\mathrm{diag}(\boldsymbol{\psi}_{bi}^{-1}) Q_b \in \mathbb{S}_+^{d_b}.$$

The scalar effective count matches G_{bi}/n to the EC-derived V_{bi}^{cond} in Frobenius inner product $\langle U, W \rangle_F = \mathrm{tr}(U'W)$, for $U, W \in \mathbb{R}^{d_b \times d_b}$:

$$n_{bi}^{\mathrm{eff}} = \min\left\{ N_{bi}^{\mathrm{gene}}, \frac{\langle G_{bi}, G_{bi} \rangle_F}{\langle G_{bi}, V_{bi}^{\mathrm{cond}} \rangle_F} \right\} \in \mathbb{R}_+,$$

where $N_{bi}^{\mathrm{gene}} \in \mathbb{N}_0$ is the gene UMI count. Fractional path counts are $\boldsymbol{x}_{bi} = n_{bi}^{\mathrm{eff}} \boldsymbol{\psi}_{bi} \in \mathbb{R}_+^{S_b}$, with $\mathbf{1}'_{S_b} \boldsymbol{x}_{bi} = n_{bi}^{\mathrm{eff}}$.

For one block, stack $n = n_b \in \mathbb{N}$ observations in a design $X \in \mathbb{R}^{n \times k}$, where $k \in \mathbb{N}$, and coefficient matrix $\boldsymbol{B}_b \in \mathbb{R}^{k \times d_b}$. The mean composition $\boldsymbol{\mu}_{bi} \in \Delta^{S_b - 1}$ is defined by reference-path logits

$$\log \frac{\mu_{bis}}{\mu_{biS_b}} = (X \boldsymbol{B}_b)_{is}, \qquad s = 1, \ldots, d_b.$$

The generalized Dirichlet–multinomial likelihood, extended to fractional counts through gamma functions, is

$$\boldsymbol{x}_{bi} \mid n_{bi}^{\mathrm{eff}} \sim \mathrm{DirichletMultinomial}\left(n_{bi}^{\mathrm{eff}}, \kappa_b \boldsymbol{\mu}_{bi} \right),$$

where $\kappa_b \in \mathbb{R}_+ \cup \{\infty\}$ is a block-level concentration. Larger κ_b means less overdispersion; the endpoint $\kappa_b = \infty$ is exactly multinomial.

For nested designs $X_0 \in \mathbb{R}^{n \times k_0}$ and $X_1 \in \mathbb{R}^{n \times k_1}$, with $k_1 \geq k_0$ and the first $k_0, k_1 \in \mathbb{N}$, and the first k_0 columns of X_1 equal to X_0, let $\widehat{\ell}_0, \widehat{\ell}_1 \in \mathbb{R}$ be maximized scalar log likelihoods. The statistic and nominal degrees of freedom are

$$\Lambda_b = 2(\widehat{\ell}_1 - \widehat{\ell}_0) \in \mathbb{R}_+, \qquad \nu_b = (k_1 - k_0) d_b \in \mathbb{N}_0.$$

The fixed-concentration analysis estimates $\widehat{\kappa}_{0b} \in \mathbb{R}_+ \cup \{\infty\}$ under X_0 and uses it under X_1. The free-concentration sensitivity estimates $\widehat{\kappa}_{1b} \in \mathbb{R}_+ \cup \{\infty\}$ under X_1. In the latter case both models contain one concentration nuisance parameter, so ν_b is unchanged. Both finite and exact multinomial endpoints are evaluated.

Gaussian mouse-replicate regression

For a condition test within one block and cell type, stack $\widehat{\boldsymbol{y}}_{bi} \in \mathbb{R}^{d_b}$ over $n \in \mathbb{N}$ mice. Let $X_b \in \mathbb{R}^{n \times k}$, $k \in \mathbb{N}$, contain an intercept and condition indicators, $\boldsymbol{B}_b^{\mathrm{GLS}} \in \mathbb{R}^{k \times d_b}$ be regression coefficients, and $\boldsymbol{\epsilon}_{bi} \in \mathbb{R}^{d_b}$ be residuals:

$$\widehat{\boldsymbol{y}}_{bi} = (X_{b,i\cdot} \boldsymbol{B}_b^{\mathrm{GLS}})' + \boldsymbol{\epsilon}_{bi}, \qquad \mathrm{Var}(\boldsymbol{\epsilon}_{bi}) = V_{bi} + \tau_b^2 I_{d_b} \in \mathbb{S}_+^{d_b}.$$

Here $X_{b,i\cdot} \in \mathbb{R}^{1 \times k}$, and $\tau_b^2 \in \mathbb{R}_+$ is residual between-mouse variance estimated by REML under the null. If $r_b \in \mathbb{N}$ non-reference condition columns are tested, their coefficients form $\boldsymbol{\beta}_b^{\mathrm{test}} \in \mathbb{R}^{r_b d_b}$ with covariance $\Sigma_b^{\mathrm{test}} \in \mathbb{S}_+^{r_b d_b}$. The Wald statistic is

$$W_b = (\boldsymbol{\beta}_b^{\mathrm{test}})' (\Sigma_b^{\mathrm{test}})^{-1} \boldsymbol{\beta}_b^{\mathrm{test}} \in \mathbb{R}_+, \qquad \nu_b^{\mathrm{GLS}} = r_b d_b.$$

Paired cell-type test and permutations

For a cell-type test, let $M_b, L_b \in \mathbb{N}$ be the numbers of represented biological mice and cell types, respectively. The null design $X_{0b} \in \{0,1\}^{n_b \times M_b}$ contains an intercept-coded set of mouse fixed effects. The alternative $X_{1b} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$ appends treatment-coded cell-type indicators. Consequently $k_0 = M_b$, $k_1 = M_b + L_b - 1$, and $\nu_b = (L_b - 1) d_b$. The tested coefficient block is $\boldsymbol{B}_b^{\mathrm{cell}} \in \mathbb{R}^{(L_b - 1) \times d_b}$, and

$$H_{0b} : \boldsymbol{B}_b^{\mathrm{cell}} = 0 \quad \text{versus} \quad H_{1b} : \boldsymbol{B}_b^{\mathrm{cell}} \neq 0.$$

Mouse effects absorb condition and every shift shared by cell types from the same mouse.

For $R = 500 \in \mathbb{N}$ permutations and $r \in \{1, \ldots, R\}$, cell-type labels are shuffled only among pseudobulks from the same mouse, producing $X_{1b}^{(r)} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$ and statistic $\Lambda_b^{(r)} \in \mathbb{R}_+$. Let $\varepsilon_b = 10^{-6} \max(1, |\Lambda_b|) \in \mathbb{R}_+$. The finite-permutation p-value is

$$p_b = \frac{1 + \sum_{r=1}^{R} \mathbb{1}\{\Lambda_b^{(r)} \geq \Lambda_b - \varepsilon_b\}}{R + 1} \in (0, 1], \qquad R = 500.$$

The null coefficients and $\widehat{\kappa}_{0b}$ are reused, but each permuted alternative is optimized. The tolerance counts numerically equal likelihood optima as ties. Benjamini–Hochberg correction is applied across canonical partitions with at least 30 pseudobulks.

Joint condition tests

ACAT combines $J \in \mathbb{N}$ component p-values $p_j \in (0,1)$, $j = 1, \ldots, J$, into the scalar

$$p_{\mathrm{ACAT}} = \frac{1}{2} - \frac{1}{\pi} \arctan\left[\frac{1}{J} \sum_{j=1}^{J} \tan\{\pi(1/2 - p_j)\} \right] \in (0, 1).$$

One level combines cell-type-specific p-values for a path partition; a second combines distinct partitions within a gene.

The shared-effect Gaussian model for block b uses $M \in \mathbb{N}$ mice indexed by $m \in \{1,\dots,M\}$, and $L_m \in \mathbb{N}$ cell types for mouse m, indexed by $c \in \{1,\dots,L_m\}$. Its response is $\boldsymbol{y}_{bmc} \in \mathbb{R}^{d_b}$, design row $X_{bmc} \in \mathbb{R}^{1\times k}$, where $k \in \mathbb{N}$, coefficient matrix $\boldsymbol{B}_b \in \mathbb{R}^{k\times d_b}$, mouse random effect $\boldsymbol{u}_{bm} \in \mathbb{R}^{d_b}$, and residual $\boldsymbol{e}_{bmc} \in \mathbb{R}^{d_b}$:

$$\boldsymbol{y}_{bmc} = (X_{bmc}\boldsymbol{B}_b)' + \boldsymbol{u}_{bm} + \boldsymbol{e}_{bmc}.$$

The random terms satisfy $\mathrm{E}(\boldsymbol{u}_{bm}) = \mathrm{E}(\boldsymbol{e}_{bmc}) = \boldsymbol{0}_{d_b}$, $\mathrm{Var}(\boldsymbol{u}_{bm}) = \tau^2_{\mathrm{mouse}} I_{d_b}$, and $\mathrm{Var}(\boldsymbol{e}_{bmc}) = V_{bmc} + \tau^2_{\mathrm{resid}} I_{d_b}$. They are independent across mice, and residuals are independent across cell types conditional on $\boldsymbol{u}_{bm}$. Hence, for $c \neq c'$,

$$\mathrm{Cov}(\boldsymbol{y}_{bmc}, \boldsymbol{y}_{bmc'}) = \tau^2_{\mathrm{mouse}} I_{d_b}, \qquad \mathrm{Var}(\boldsymbol{y}_{bmc}) = V_{bmc} + (\tau^2_{\mathrm{mouse}} + \tau^2_{\mathrm{resid}}) I_{d_b},$$

where $V_{bmc} \in \mathbb{S}^{d_b}_{+}$, and $\tau^2_{\mathrm{mouse}}, \tau^2_{\mathrm{resid}} \in \mathbb{R}_+$ are shared-mouse and residual variance components estimated by two-component REML under the no-condition model. Condition labels are permuted once per mouse and applied to all its cell types.

Multinomial-logit GLMM

The formal generalized linear mixed model uses the same shared-condition design as the joint Gaussian model but retains the multinomial response. Let $\widetilde{\boldsymbol{x}}_{bmc} \in \mathbb{N}_0^{S_b}$ be integer path counts with $\widetilde{n}_{bmc} = \boldsymbol{1}'_{S_b}\widetilde{\boldsymbol{x}}_{bmc} \in \mathbb{N}$. They are obtained from the fractional effective counts $\boldsymbol{x}_{bmc}$ by rounding n^{eff}_{bmc} to the nearest positive integer and distributing the remaining molecules to paths by largest fractional remainder. This makes the conditional response a proper multinomial while changing each effective total by at most one.

For design row $X_{bmc} \in \mathbb{R}^{1\times k}$, fixed coefficients $\boldsymbol{B}_b \in \mathbb{R}^{k\times d_b}$, and mouse random intercept $\boldsymbol{u}_{bm} \in \mathbb{R}^{d_b}$, define

$$\boldsymbol{\xi}_{bmc} = X_{bmc}\boldsymbol{B}_b + \boldsymbol{u}'_{bm} \in \mathbb{R}^{1\times d_b}, \qquad \boldsymbol{\mu}_{bmc} = \mathrm{softmax}\{(\boldsymbol{\xi}_{bmc}, 0)\} \in \Delta^{S_b - 1}.$$

The final path is the reference category. The model is

$$\widetilde{\boldsymbol{x}}_{bmc} \mid \boldsymbol{u}_{bm} \sim \mathrm{Multinomial}(\widetilde{n}_{bmc}, \boldsymbol{\mu}_{bmc}), \qquad \boldsymbol{u}_{bm} \sim N(\boldsymbol{0}_{d_b}, \sigma_b^2 I_{d_b}),$$

where $\sigma_b^2 \in \mathbb{R}_+$ is re-estimated under both null and alternative. An isotropic random-effect covariance is used because some blocks have few mice; an unstructured $d_b \times d_b$ covariance would introduce $d_b(d_b+1)/2$ variance parameters per block.

Let $\ell_{bm}(\boldsymbol{u}; \boldsymbol{B}_b, \sigma_b) \in \mathbb{R}$ be mouse m's conditional multinomial log likelihood plus its Gaussian log density, and let

$$\widehat{\boldsymbol{u}}_{bm} = \arg\max_{\boldsymbol{u}\in\mathbb{R}^{d_b}} \ell_{bm}(\boldsymbol{u}; \boldsymbol{B}_b, \sigma_b), \qquad H_{bm} = -\left.\frac{\partial^2 \ell_{bm}}{\partial \boldsymbol{u}\,\partial \boldsymbol{u}'}\right|_{\widehat{\boldsymbol{u}}_{bm}} \in \mathbb{S}^{d_b}_{+}.$$

The Laplace log marginal likelihood, up to constants common to nested models, is

$$\ell_b^{\mathrm{Lap}}(\boldsymbol{B}_b, \sigma_b) = \sum_{m=1}^{M}\left[\ell_{bm}(\widehat{\boldsymbol{u}}_{bm}; \boldsymbol{B}_b, \sigma_b) + \frac{d_b}{2}\log(2\pi) - \frac{1}{2}\log|H_{bm}|\right] \in \mathbb{R}.$$

The implementation differentiates this approximation through 30 damped Newton updates for every $\widehat{\boldsymbol{u}}_{bm}$. The null contains cell-type intercepts; the alternative appends r_b condition columns. The likelihood-ratio statistic has nominal degrees of freedom $r_b d_b$.

Additional EC likelihoods and inference approximations

The path-count GLMM above treats estimated effective path counts as its response. The following restatement of the observed-EC model supplies context for the additional Dirichlet–multinomial, Laplace, CAVI, Monte Carlo, and Rényi inference comparisons. For gene g, let $T_g \in \mathbb{N}$ be its number of supported isoforms, $d_g = T_g - 1$, $I_g \in \mathbb{N}$ the number of retained pseudobulks, $M_g \in \mathbb{N}$ the number of mice, and $P = 2$ the primer count. Primer p has $K_{gp} \in \mathbb{N}$ gene-assigned ECs. Define the observed counts $\boldsymbol{c}_{gip} \in \mathbb{N}_0^{K_{gp}}$, total $n_{gip} = \mathbf{1}'_{K_{gp}}\boldsymbol{c}_{gip} \in \mathbb{N}_0$, and fixed compatibility or conditional-weight matrix $A_{gp} \in \mathbb{R}_+^{K_{gp}\times T_g}$. Unsupported annotated isoforms and ECs not uniquely assigned to g are excluded.

Let $X_g \in \mathbb{R}^{I_g\times q}$ be the fixed-effect design, $B_g \in \mathbb{R}^{q\times d_g}$ its coefficient matrix, and $u_{gm} \in \mathbb{R}^{d_g}$ mouse m's random isoform-logit intercept. With mouse index $m(i) \in \{1,\ldots,M_g\}$, the reference-isoform parameterization is

$$\eta_{gi} = \{X_{gi}B_g + u'_{g,m(i)}, 0\} \in \mathbb{R}^{T_g}, \qquad u_{gm} \sim N(0, \sigma_g^2 I_{d_g}).$$

The probability of EC k from primer p is

$$q_{gipk}(\eta_{gi}) = \frac{\sum_{t=1}^{T_g} A_{gpkt}\exp(\eta_{git})}{\sum_{h=1}^{K_{gp}}\sum_{t=1}^{T_g} A_{gpht}\exp(\eta_{git})}, \qquad \boldsymbol{q}_{gip} \in \Delta^{K_{gp}-1}.$$

Thus the two primers share B_g and u_{gm}, but retain distinct EC maps and positional-coverage weights. The conditional model is either

$$\boldsymbol{c}_{gip} \mid u_{g,m(i)} \sim \mathrm{Multinomial}(n_{gip}, \boldsymbol{q}_{gip})$$

or

$$\boldsymbol{c}_{gip} \mid u_{g,m(i)} \sim \mathrm{DirichletMultinomial}(n_{gip}, \kappa_g \boldsymbol{q}_{gip}), \qquad \kappa_g \in \mathbb{R}_+.$$

The latter adds pseudobulk-level EC overdispersion. It is a direct Dirichlet–multinomial over EC categories, not a Dirichlet model over latent isoform allocations.

An alternative overdispersion model retains the multinomial observation but adds an isotropic observation-level logistic-normal effect. For pseudobulk i, let $e_{gi} \in \mathbb{R}^{d_g}$ and replace the linear predictor by

$$\eta_{gi} = \{X_{gi}B_g + u'_{g,m(i)} + e'_{gi}, 0\}, \qquad e_{gi} \sim N(0, \tau_g^2 I_{d_g}), \quad \tau_g \in \mathbb{R}_+.$$

The same e_{gi} enters both primer likelihoods, because it perturbs the underlying isoform composition rather than each primer's EC probabilities independently. Consequently this model allows correlated extra-multinomial variation across the two primer halves. It is a logistic-normal analogue of the Dirichlet–multinomial, not an identical distribution: its dispersion is isotropic in reference-logit coordinates, whereas the Dirichlet–multinomial has one concentration on the EC simplex.

Laplace inference maximizes the joint conditional density over $u_g \in \mathbb{R}^{M_g\times d_g}$. Its negative Hessian satisfies $H_g \in \mathbb{S}_+^{M_g d_g}$, and the marginal approximation adds the correction $-\frac{1}{2}\log|H_g|$. The outer fit optimizes B_g, σ_g, and, for the overdispersed model, κ_g.

For outer parameter vector $\boldsymbol{\theta}_g \in \mathbb{R}^{p_g}$, cluster mode $\widehat{\boldsymbol{u}}_{gm} \in \mathbb{R}^{d_g}$, mode score $\boldsymbol{s}_{gm} = \partial\ell_{gm}/\partial\boldsymbol{u}_{gm} \in \mathbb{R}^{d_g}$, and positive mode Hessian $H_{gm} = -\partial\boldsymbol{s}_{gm}/\partial\boldsymbol{u}_{gm} \in \mathbb{S}_{++}^{d_g}$, the fitted mode satisfies $\boldsymbol{s}_{gm}(\widehat{\boldsymbol{u}}_{gm}, \boldsymbol{\theta}_g) = \mathbf{0}_{d_g}$. Rather than retaining and differentiating through every Newton update, the optimized implementation differentiates this stationarity equation implicitly,

$$\frac{\partial\widehat{\boldsymbol{u}}_{gm}}{\partial\boldsymbol{\theta}_g} = H_{gm}^{-1}\frac{\partial\boldsymbol{s}_{gm}}{\partial\boldsymbol{\theta}_g} \in \mathbb{R}^{d_g\times p_g}.$$

For a reverse-mode cotangent $\boldsymbol{v}_{gm} \in \mathbb{R}^{d_g}$, it solves one $d_g \times d_g$ system with H_{gm} and applies the score vector–Jacobian product. This gives the derivative of the same Laplace objective at a converged mode without backpropagating through the 12 Newton updates. Fits that fail the joint outer-gradient and mode-score criteria receive a 30-update continuation. Compiled objectives are reused only among contiguous tests with the same gene, pseudobulk set, and tensor shape; counts and block-specific fixed-effect tensors are dynamic arguments, and the cache is released before the next group.

Nonbootstrap Laplace inference

The block test uses the same fixed-effect tensors as the primary variational fit. Write the null tensor as $Z_{0b} \in \mathbb{R}^{I_g \times d_g \times p_{0b}}$ and the alternative as $Z_{1b} \in \mathbb{R}^{I_g \times d_g \times p_{1b}}$, where the first p_{0b} slices of Z_{1b} equal Z_{0b}. The remaining $q_b = p_{1b} - p_{0b} = r_b d_b^{\text{path}}$ coefficients are exactly the tested block-path interactions. For coefficient vector $\boldsymbol{\beta}_b \in \mathbb{R}^{p_{jb}}$, the fixed nonreference-isoform logits for pseudobulk i are

$$\boldsymbol{\eta}_{gbi}^{\text{fix}} = Z_{jb,i,\cdot,\cdot}\boldsymbol{\beta}_b \in \mathbb{R}^{d_g}, \qquad j \in \{0, 1\}.$$

Thus the likelihood still contains every supported gene isoform and EC; only the tested fixed-effect subspace is block specific.

Let $\widehat{\ell}_{0b}^{\text{Lap}}$ and $\widehat{\ell}_{1b}^{\text{Lap}}$ be the maximized Laplace marginal log likelihoods, including re-estimation of all shared variance parameters. The nonbootstrap likelihood-ratio statistic and nominal reference distribution are

$$D_b = 2\{\widehat{\ell}_{1b}^{\text{Lap}} - \widehat{\ell}_{0b}^{\text{Lap}}\}, \qquad D_b \mathrel{\dot{\sim}} \chi^2_{q_b}.$$

The alternative starts from the null estimate with its q_b added coefficients set to zero. A BIC approximation to the alternative-versus-null Bayes factor uses the number M_g of independent mice rather than the number of pseudobulks:

$$\log \widehat{\text{BF}}_{10,b}^{\text{BIC}} = \frac{1}{2}\{D_b - q_b \log M_g\} \in \mathbb{R}.$$

This is not a prior-integrated Bayes factor. Both calculations require only the null and alternative fits, compared with one null and one alternative fit for every parametric-bootstrap replicate. Their calibration can fail when the number of mice is small, a variance estimate lies on its boundary, or the Laplace outer optimizer is inaccurate. They are therefore compared with the completed mouse bootstrap analysis before being used as a replacement for empirical calibration.

For variational inference, use the diagonal Gaussian family

$$Q(u_g) = \prod_{m=1}^{M_g} \prod_{j=1}^{d_g} N(u_{gmj}; \mu_{gmj}, s_{gmj}^2), \quad \mu_g, s_g \in \mathbb{R}^{M_g \times d_g}, \quad s_{gmj} > 0.$$

For the multinomial likelihood, latent EC-to-isoform responsibilities $r_{gipkt} \in [0, 1]$, with $\sum_t r_{gipkt} = 1$ on the support of A_{gpkt}, give the optimized lower bound

$$E_Q\left[\log \sum_t A_{gpkt} e^{\eta_{git}}\right] \geq \log \sum_t A_{gpkt} e^{E_Q \eta_{git}}.$$

The normalizer needs an upper bound. For one observation and primer, let $m_{gipt} = E_Q \eta_{git} + \log \sum_k A_{gpkt}$, $v_{gipt} = \text{Var}_Q(\eta_{git})$, and $a_{gip} \in \Delta^{T_g - 1}$. The tilted softmax bound of Knowles and

Minka is

$$E_Q \log \sum_{t=1}^{T_g} e^{\eta_{git} + \log \sum_k A_{gpkt}} \leq \frac{1}{2} \sum_t a_{gipt}^2 v_{gipt} + \log \sum_t \exp\left\{ m_{gipt} + \frac{1 - 2a_{gipt}}{2} v_{gipt} \right\},$$

where the local coordinate satisfies

$$a_{gip} = \mathrm{softmax}\{ m_{gip} + \tfrac{1}{2}(1 - 2a_{gip}) \odot v_{gip} \}.$$

Alternating the analytic responsibilities, this local fixed point, and the global Gaussian parameters maximizes a tractable lower bound on the multinomial evidence. The cited method is non-conjugate variational message passing rather than conjugate CAVI; here "tilted CAVI" refers only to this block-coordinate implementation.

The full-covariance extension factorizes only across mice. If mouse m has $q_{gm} \in \mathbb{N}$ observed pseudobulks, define its complete latent block

$$z_{gm} = (u'_{gm}, e'_{gi_1}, \ldots, e'_{gi_{q_{gm}}})' \in \mathbb{R}^{L_{gm}}, \qquad L_{gm} = (q_{gm} + 1) d_g,$$

where the e components are omitted in the model without observation noise. Its prior covariance is diagonal,

$$P_{gm} = \mathrm{diag}(\sigma_g^2 I_{d_g}, \tau_g^2 I_{q_{gm} d_g}) \in \mathbb{S}_+^{L_{gm}},$$

but its posterior approximation is unrestricted within mouse,

$$Q(z_g) = \prod_{m=1}^{M_g} N(z_{gm}; \mu_{gm}, \Sigma_{gm}), \quad \mu_{gm} \in \mathbb{R}^{L_{gm}}, \quad \Sigma_{gm} \in \mathbb{S}_{++}^{L_{gm}}.$$

Thus full covariance describes posterior dependence among isoform logits and between a shared mouse effect and its pseudobulk residuals; it does not add an unstructured covariance parameter to the generative model.

For a block-specific cell-type test, the EC likelihood still contains all T_g supported isoforms of gene g; the response is not reduced to path counts. Let $J_b \in \mathbb{N}$ be the number of represented cell types, $Z_b \in \mathbb{R}^{I_g \times q_0}$ the one-hot condition design, and $W_b \in \mathbb{R}^{I_g \times (J_b - 1)}$ the treatment-coded cell-type design. Define $D_b \in \mathbb{R}^{T_g \times d_b}$ by assigning row t the corresponding row of Q_b when isoform t takes a represented path through block b, and assigning it $\mathbf{0}'_{d_b}$ when the isoform is outside the block map. Center the columns of D_b, and let $N_b^{\mathrm{full}} \in \mathbb{R}^{T_g \times (T_g - S_b)}$ be an orthonormal basis for the subspace orthogonal to both its columns and $\mathbf{1}_{T_g}$. In reference-isoform coordinates, define

$$H_b = D_{b,1:(T_g-1),:} - \mathbf{1}_{T_g-1} D_{b,T_g,:} \in \mathbb{R}^{d_g \times d_b}, \qquad N_b = N^{\mathrm{full}}_{b,1:(T_g-1),:} - \mathbf{1}_{T_g-1} N^{\mathrm{full}}_{b,T_g,:} \in \mathbb{R}^{d_g \times (T_g - S_b)}.$$

The free-logit predictor is

$$\eta_{gi,-} = Z_{bi} B_{g0} + \sum_{j=1}^{J_b - 1} W_{bij} (N_b \boldsymbol{\alpha}_{bj} + H_b \boldsymbol{\gamma}_{bj}) + u_{g,m(i)} + e_{gi} \in \mathbb{R}^{d_g},$$

where $B_{g0} \in \mathbb{R}^{q_0 \times d_g}$ contains unrestricted condition effects, $\boldsymbol{\alpha}_{bj} \in \mathbb{R}^{T_g - S_b}$ contains cell-type nuisance effects orthogonal to the tested block directions, and $\boldsymbol{\gamma}_{bj} \in \mathbb{R}^{d_b}$ is cell type j's block-path contrast. The e_{gi} term is present only in the logistic-normal sensitivity model. The null estimates every $\boldsymbol{\alpha}_{bj}$ but sets all $\boldsymbol{\gamma}_{bj} = \mathbf{0}_{d_b}$; the alternative has $(J_b - 1) d_b$ additional coefficients. The combined columns of N_b and H_b span all d_g isoform-logit directions, so the alternative permits unrestricted cell-type isoform composition while the null absorbs every direction orthogonal to the tested path contrast. This construction makes the EC GLMM a per-block test while preserving gene-wide ambiguity and nuisance mass.

For pseudobulk i, let $R_{gi} \in \{0,1\}^{d_g \times L_{g,m(i)}}$ select and sum its mouse and pseudobulk effects. Its free-logit covariance is $R_{gi}\Sigma_{g,m(i)}R'_{gi} \in \mathbb{S}_+^{d_g}$; append a zero row and column for the reference isoform to obtain $V_{gi} \in \mathbb{S}_+^{T_g}$. The correlated tilted bound replaces the variance vector in the diagonal expression. For $m \in \mathbb{R}^{T_g}$, $V \in \mathbb{S}_+^{T_g}$, and $a \in \Delta^{T_g - 1}$,

$$E \log \sum_t e^{\eta_t} \le \frac{1}{2} a'Va + \log \sum_t \exp\{m_t + \tfrac{1}{2}V_{tt} - (Va)_t\}, \qquad a = \operatorname{softmax}\{m + \tfrac{1}{2}\operatorname{diag}(V) - Va\}.$$

The Gaussian KL term is analytic:

$$\mathrm{KL}(Q_m \| P_m) = \frac{1}{2}\left[\operatorname{tr}(P_m^{-1}\Sigma_m) + \mu_m' P_m^{-1}\mu_m - L_{gm} + \log|P_m| - \log|\Sigma_m|\right].$$

Each positive-definite Σ_m is represented by an unconstrained lower Cholesky factor with exponentiated diagonal. The full fit warm-starts from a diagonal variational posterior by copying its means and marginal variances and setting new off-diagonal entries to zero.

The direct EC Dirichlet–multinomial contains $\log\Gamma\{\kappa_g q_{gipk}(u) + c_{gipk}\}$, so the tilted softmax bound does not yield coordinate updates for it. Both likelihood families are therefore also fit with reparameterized Monte Carlo VI. If $w(u) = p(c,u)/Q(u)$, the optimized objective is the ELBO

$$\mathcal{L}_1 = E_Q \log w(u)$$

or the variational Rényi bound of order $\alpha \in (0,1)$,

$$\mathcal{L}_\alpha = \frac{1}{1-\alpha}\log E_Q\{w(u)^{1-\alpha}\}.$$

The comparison uses $\alpha = 0.5$, antithetic Gaussian draws, and the same diagonal or full Gaussian family. Conditional on B_g, σ_g, κ_g, both the likelihood and Q factor by mouse. Writing the corresponding ratio as $w_m(u_{gm})$, the implementation evaluates the equivalent expression

$$\mathcal{L}_\alpha = \sum_{m=1}^{M_g} \frac{1}{1-\alpha}\log E_{Q_m}\{w_m(u_{gm})^{1-\alpha}\}.$$

This avoids the exponential importance-weight degeneracy caused by forming one product weight over all mice. Rényi VI is not called CAVI because its importance-weighted objective has no coordinate update under this model. For the full posterior with observation-level noise, directly optimizing a fixed set of Monte Carlo draws is ill-conditioned: the Cholesky factor can overfit sample correlations when L_{gm} is not small relative to the draw count. This variant is therefore fit with the analytic tilted objective; fresh antithetic draws evaluate its exact-likelihood ELBO and Rényi bound.

The block statistic is twice the optimized alternative-minus-null variational objective. Its reference distribution is empirical rather than χ^2, because it compares optimized evidence bounds and can inherit approximation error. A synchronized label permutation is invalid here because the block null permits real cell-type nuisance effects in orthogonal isoform directions. Instead, a parametric bootstrap simulates mouse effects and optional pseudobulk effects from the fitted block null, generates paired-primer EC counts with the observed primer-specific pseudobulk totals, and refits both nested models. Each observed statistic is compared with a leave-block-out pool of bootstrap statistics matched by likelihood family and tested dimension $(J_b - 1)d_b$. Tested dimensions represented by fewer than 20 blocks are combined into a rare-dimension stratum within the same likelihood. Matching gene-depth quartiles fragmented the empirical tail enough to limit BH resolution in the genome-wide analysis. Depth is therefore an independent calibration audit rather than a pooling variable. With null pool $\mathcal{N}_{-b}$,

$$p_b = \frac{1 + \sum_{z \in \mathcal{N}_{-b}} \mathbf{1}\{z \geq T_b\}}{1 + |\mathcal{N}_{-b}|}.$$

Every bootstrap statistic is calibrated by the same leave-block-out rule. Benjamini–Hochberg values are reported for a likelihood only if these calibrated null p-values reject between 2.5% and 7.5% at level 0.05, both in aggregate and separately within each gene-depth quartile.

Liang–Zeger multinomial GEE

GEE specifies only the marginal mean and first two moments. It can therefore use the fractional effective counts directly. Define $\boldsymbol{p}_{bmc} = \boldsymbol{x}_{bmc}/n^{\text{eff}}_{bmc} \in \Delta^{S_b - 1}$, and let $\boldsymbol{p}_{bmc,-} \in \mathbb{R}^{d_b}$ omit the reference path. Its marginal mean uses the same reference-logit regression,

$$\boldsymbol{\mu}_{bmc} = \text{softmax}\{(X_{bmc}\boldsymbol{B}_b, 0)\}, \qquad G_{bmc} = \text{diag}(\boldsymbol{\mu}_{bmc,-}) - \boldsymbol{\mu}_{bmc,-}\boldsymbol{\mu}'_{bmc,-} \in \mathbb{S}^{d_b}_{+}.$$

The working covariance of one proportion vector is $A_{bmc} = G_{bmc}/n^{\text{eff}}_{bmc} \in \mathbb{S}^{d_b}_{+}$. Let $L_{bmc} \in \mathbb{R}^{d_b \times d_b}$ be its symmetric square root, and let

$$D_{bmc} = \frac{\partial \boldsymbol{\mu}_{bmc,-}}{\partial \, \text{vec}(\boldsymbol{B}_b)'} = [X_{bmc,1}G_{bmc} \ \cdots \ X_{bmc,k}G_{bmc}] \in \mathbb{R}^{d_b \times k d_b}.$$

For mouse m, stack its $q_m \in \mathbb{N}$ observed cell types into $\boldsymbol{p}_{bm,-} \in \mathbb{R}^{q_m d_b}$, mean $\boldsymbol{\mu}_{bm,-} \in \mathbb{R}^{q_m d_b}$, and derivative $D_{bm} \in \mathbb{R}^{q_m d_b \times k d_b}$. With $L_{bm} = \text{blockdiag}_c(L_{bmc})$ and exchangeable $R_m(\alpha_b) \in \mathbb{R}^{q_m \times q_m}$, the Liang–Zeger working covariance is

$$W_{bm} = \phi_b L_{bm}\{R_m(\alpha_b) \otimes I_{d_b}\}L'_{bm} \in \mathbb{S}^{q_m d_b}_{+}.$$

The scale $\phi_b \in \mathbb{R}_+$ and working correlation $\alpha_b \in (-1/(q_{\max} - 1), 1)$ are updated from whitened Pearson residuals. Coefficients solve

$$\sum_{m=1}^{M} D'_{bm} W^{-1}_{bm} (\boldsymbol{p}_{bm,-} - \boldsymbol{\mu}_{bm,-}) = \mathbf{0}_{k d_b}.$$

If $\boldsymbol{U}_{bm} = D'_{bm} W^{-1}_{bm} (\boldsymbol{p}_{bm,-} - \boldsymbol{\mu}_{bm,-}) \in \mathbb{R}^{k d_b}$ and $\mathcal{H}_b = \sum_m D'_{bm} W^{-1}_{bm} D_{bm} \in \mathbb{S}^{k d_b}_{+}$, the canonical cluster-robust covariance is

$$\widehat{\text{Var}}\{\text{vec}(\widehat{\boldsymbol{B}}_b)\} = \mathcal{H}^{+}_b \left(\sum_{m=1}^{M} \boldsymbol{U}_{bm}\boldsymbol{U}'_{bm} \right) \mathcal{H}^{+}_b \in \mathbb{S}^{k d_b}_{+}.$$

The condition coefficients are tested by a robust Wald statistic. No small-sample sandwich correction is imposed; synchronized mouse-level permutations assess whether the canonical large-M reference is adequate.

Matched power benchmark

Power is evaluated only for partitions represented by exactly two conditions. For replicate r, condition labels are first permuted among mice. Let $z_{bmc}^{(r)} \in \{0,1\}$ be the resulting non-reference condition indicator and let $\boldsymbol{v}_b \in \mathbb{R}^{d_b}$ be a reproducible random unit vector. For ILR effect $\gamma \in \{0.25, 0.5\}$, define

$$\boldsymbol{y}_{bmc}^{(r,\gamma)} = \boldsymbol{y}_{bmc} + \gamma z_{bmc}^{(r)} \boldsymbol{v}_b, \qquad \boldsymbol{\psi}_{bmc}^{(r,\gamma)} = \operatorname{softmax}\{Q_b \boldsymbol{y}_{bmc}^{(r,\gamma)}\}.$$

The shifted effective counts are $n_{bmc}^{\text{eff}} \boldsymbol{\psi}_{bmc}^{(r,\gamma)}$. Thus every method sees the same imposed ILR effect, observed effective depth, cell-type pattern, and mouse clustering. For each simulated alternative, null-calibrated power uses the finite-permutation rank of its statistic among synchronized null statistics for the same block and method. This event-matched calibration preserves dimension, depth, and cluster structure.

B Supporting Algorithms

All algorithms operate independently over canonical blocks unless a joint gene-level combination is stated. "Fit" below means bounded L-BFGS with analytic gradients and a fourfold iteration-limit retry after a warm start from the first endpoint.

Algorithm 4 Construct splice blocks and path maps

Require: K_g annotated spliced isoforms with exon chains; retained set $\mathcal{I}_g$, $|\mathcal{I}_g| = T_g$
Ensure: Blocks b, maps $C_b \in \{0,1\}^{S_b \times T_g}$, and bases $Q_b \in \mathbb{R}^{S_b \times d_b}$
Split the gene at every annotated exon boundary into atomic intervals.
Merge adjacent intervals exonic in every annotated isoform into constitutive anchors.
Form internal blocks between consecutive anchors and terminal blocks using pseudo-anchors.
for all candidate blocks b **do**
  Record each annotated isoform's ordered intervening exon chain.
  Assign identical chains to one path and restrict to $\mathcal{I}_b^{\text{sp}} \subseteq \mathcal{I}_g$.
  Remap represented path labels to $\{1, \ldots, S_b\}$.
  if $S_b < 2$ **then**
    discard b
  else
    Set $(C_b)_{s_b(t),t} = 1$ for $t \in \mathcal{I}_b^{\text{sp}}$; leave nuisance columns zero.
    Construct the Helmert basis Q_b.
  end if
end for
Canonicalize the unordered isoform partition and retain one representative of equivalent blocks.

Algorithm 5 Fit one pseudobulk's local path composition

Require: Counts $\boldsymbol{c}_{bip} \in \mathbb{N}_0^{E_{bp}}$, designs $A_{bp} \in \mathbb{R}_+^{E_{bp} \times T_g}$, baseline $\boldsymbol{\theta}_{bi}^{(0)} \in \mathbb{R}_+^{T_g}$, C_b, and Q_b
Ensure: $\widehat{\boldsymbol{\theta}}_{bi}$, $\widehat{\boldsymbol{\psi}}_{bi}$, $\widehat{\boldsymbol{y}}_{bi}$, and V_{bi}^{cond}

Initialize $\boldsymbol{\delta}_{bi} \leftarrow \mathbf{0}_{d_b}$.
if $S_b = 2$ **then**
    Evaluate the fixed log-ratio grid and retain its best point.
end if
function OBJECTIVE($\boldsymbol{\delta}$)
    Form $\boldsymbol{\theta}_{bi}(\boldsymbol{\delta})$, preserving spliced mass and within-path isoform mixtures.
    return the negative sum of the $P = 2$ multinomial log likelihoods and its analytic gradient.
end function
Minimize OBJECTIVE($\boldsymbol{\delta}$) by bounded L-BFGS on $[-20, 20]^{d_b}$.
Set $\widehat{\boldsymbol{\theta}}_{bi} \leftarrow \boldsymbol{\theta}_{bi}(\widehat{\boldsymbol{\delta}}_{bi})$.
Set $\widehat{\boldsymbol{\psi}}_{bi} \leftarrow C_b \widehat{\boldsymbol{\theta}}_{bi} / (\mathbf{1}'_{S_b} C_b \widehat{\boldsymbol{\theta}}_{bi})$.
Set $\widehat{\boldsymbol{y}}_{bi} \leftarrow Q'_b \log \widehat{\boldsymbol{\psi}}_{bi}$.
Compute $I_{\delta,bi}$ by summing primer expected information.
Apply the eigenvalue/null-space check to $(I_{\delta,bi}, I_{d_b})$; if identifiable, set $V_{bi}^{\mathrm{cond}} = I_{\delta,bi}^{-1}$, otherwise set it to infinity.

Algorithm 6 Compute full-isoform nuisance (profile) covariance

Require: $\widehat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}_+^{T_g}$, A_{bp}, N_{bip}, C_b, Q_b, relative threshold $r_{\mathrm{eig}} = 10^{-8}$, and null tolerance $a_{\mathrm{null}} = 10^{-9}$
Ensure: $V_{bi}^{\mathrm{prof}} \in \mathbb{S}_+^{d_b}$ and an identifiability flag

for $p = 1, \ldots, P$ **do**
    Compute $\widehat{\boldsymbol{q}}_{bip}$, $\widehat{Z}_{bip}$, and $I_{\eta,bip} \in \mathbb{S}_+^{T_g}$.
end for
Set $I_{\eta,bi} \leftarrow \sum_p I_{\eta,bip}$.
Compute path masses a_{bis}, spliced mass $\omega_{bi}^{\mathrm{sp}}$, and $R_{bi} = \partial \log \boldsymbol{\psi}_{bi} / \partial \boldsymbol{\eta}'_{bi}$.
Set $J_{bi} \leftarrow Q'_b R_{bi}$.
Eigendecompose $I_{\eta,bi} = U_{bi} \operatorname{diag}(\boldsymbol{\lambda}_{bi}) U'_{bi}$.
Put t in $\mathcal{P}_{bi}$ when $\lambda_{bit} > r_{\mathrm{eig}} \max_u |\lambda_{biu}|$, and put all other indices in $\mathcal{N}_{bi}$.
if $\|J_{bi} U_{bi,0}\|_F > a_{\mathrm{null}} \max\{\|J_{bi}\|_F, 1\}$ **then**
    Set $V_{bi}^{\mathrm{prof}} \leftarrow \infty_{d_b \times d_b}$ and mark the response unidentified.
else
    Set $V_{bi}^{\mathrm{prof}} \leftarrow J_{bi} U_{bi,+} \operatorname{diag}(\boldsymbol{\lambda}_{bi,\mathcal{P}}^{-1}) U'_{bi,+} J'_{bi}$.
    Mark the response identifiable.
end if

Algorithm 7 Fit a cell-resolved pseudobulk path composition

Require: Cell EC vectors $\boldsymbol{c}_{bihp}$, baselines $\boldsymbol{\theta}_{bih}^{(0)}$, weights w_{bih}, and A_{bp}, C_b, Q_b
Ensure: One pseudobulk response $\widehat{\boldsymbol{y}}_{bi}$ and covariance V_{bi}

```
Allocate one shared δ_bi ∈ ℝ^{d_b}.
for all cells h and primers p do
    Perturb the cell's baseline, normalize its EC probabilities, and add its multinomial log likelihood and gradient.
end for
Optimize the summed objective over δ_bi.
Compute each fitted ψ̂_bih and their normalized w_bih-weighted mean ψ̂_bi.
Sum cell-by-primer Fisher matrices and propagate their inverse through the weighted composition response to obtain V_bi.
Set ŷ_bi = Q'_b log ψ̂_bi.
```

Algorithm 8 Gaussian condition or paired cell-type test

Require: Responses $\widehat{\boldsymbol{y}}_{bi}$, covariances V_{bi}, mouse, condition, and cell-type labels; test type
Ensure: One Wald p-value for block b

```
if testing condition within a cell type then
    Retain conditions represented by at least three mice.
    Build full design X_b ∈ ℝ^{n×k} and reduced design X_0b ∈ ℝ^{n×(k−r_b)}.
else
    Build mouse-effect design X_0b ∈ {0,1}^{n_b×M_b}.
    Append L_b − 1 cell-type columns to obtain X_1b ∈ ℝ^{n_b×(M_b+L_b−1)}.
end if
Require full column rank in both designs.
Estimate τ_b^2 by REML under the reduced design using V_bi + τ_b^2 I_{d_b}.
Hold τ̂_b^2 fixed and fit the full design by GLS.
Stack the tested coefficients and compute their Wald statistic W_b.
Compare W_b with the applicable chi-squared reference.
Calibrate by mouse-level condition permutations or within-mouse cell-type permutations, as appropriate.
```

Algorithm 9 Paired compositional cell-type test

Require: $\widehat{\boldsymbol{\psi}}_{bi}$, V_{bi}^{cond}, mouse and cell-type labels, and $R = 500$
Ensure: One exact permutation p-value p_b per canonical block

Keep converged, identifiable estimates satisfying the path floor and the three-mouse-per-cell-type requirement.
Convert each $(\widehat{\boldsymbol{\psi}}_{bi}, V_{bi}^{\mathrm{cond}})$ to n_{bi}^{eff} and $\boldsymbol{x}_{bi}$.
Build X_{0b} from mouse effects and X_{1b} by appending cell-type effects.
Fit the null coefficients and $\widehat{\kappa}_{0b}$.
Fit the observed alternative and compute Λ_b.
if using fixed concentration **then**
    Set $\kappa_{1b} \leftarrow \widehat{\kappa}_{0b}$.
else
    Optimize $\widehat{\kappa}_{1b}$ and compare the finite optimum with the exact multinomial endpoint.
end if
for $r = 1, \ldots, R$ **do**
    Shuffle cell-type labels within each mouse to form $X_{1b}^{(r)}$.
    Reuse the null fit, refit the alternative, and store $\Lambda_b^{(r)}$.
end for
Compute the tolerance-aware finite-permutation rank p_b.
Require $n_b \geq 30$, then apply BH across canonical blocks.

Algorithm 10 Liang–Zeger multinomial GEE condition test

Require: Fractional counts $\boldsymbol{x}_{bmc}$, design X_b, mouse labels, and tested condition columns
Ensure: Robust Wald statistic and p-value

Initialize $\boldsymbol{B}_b$ from independent multinomial regression and set $\alpha_b \leftarrow 0$, $\phi_b \leftarrow 1$.
repeat
    Compute $\boldsymbol{\mu}_{bmc}$, G_{bmc}, A_{bmc}, and D_{bmc} for every observation.
    Whiten residuals by $A_{bmc}^{-1/2}$.
    Update ϕ_b from squared Pearson residuals and α_b from within-mouse Pearson cross-products.
    Construct every W_{bm}, quasi-score $\boldsymbol{U}_{bm}$, and sensitivity matrix $\mathcal{H}_b$.
    Update $\mathrm{vec}(\boldsymbol{B}_b) \leftarrow \mathrm{vec}(\boldsymbol{B}_b) + \mathcal{H}_b^{+} \sum_m \boldsymbol{U}_{bm}$.
until the maximum coefficient update is at most 10^{-8}
Form the cluster sandwich covariance $\mathcal{H}_b^{+}(\sum_m \boldsymbol{U}_{bm}\boldsymbol{U}_{bm}')\mathcal{H}_b^{+}$.
Compute the robust Wald test for all condition coefficients.
Permute condition once per mouse and repeat the complete GEE fit.

Algorithm 11 Laplace multinomial GLMM condition test

Require: Effective counts, nested designs, mouse labels, and random-effect dimension d_b
Ensure: Likelihood-ratio statistic and p-value

Integerize each effective count vector by largest remainder.
for all null and alternative models **do**
    Initialize fixed coefficients from multinomial regression and initialize σ_b.
    repeat
        for all mice m **do**
            Find $\widehat{\boldsymbol{u}}_{bm}$ with 30 damped Newton updates.
            Compute random-effect Hessian H_{bm} and its log determinant.
        end for
        Sum the Laplace log marginal likelihood and differentiate through the Newton updates.
        Update $\boldsymbol{B}_b$ and $\log \sigma_b$ by bounded L-BFGS.
    until the outer gradient tolerance or iteration limit is reached
end for
Compute twice the null-to-alternative Laplace log-likelihood gain.
Compare with $\chi^2_{r_b d_b}$.
For permutations, reuse the fitted null and refit each permuted alternative.

Algorithm 12 Nonbootstrap paired-EC Laplace block test

Require: Paired EC counts, compatibility matrices, nested tensors Z_{0b}, Z_{1b}, and mouse indices
Ensure: LRT p-value and BIC Bayes-factor approximation

Fit the null by maximizing its Laplace marginal log likelihood.
Append q_b zero coefficients to the fitted null parameters.
Warm-start and fit the alternative, re-estimating shared variance parameters.
Set $D_b \leftarrow 2\max\{0, \widehat{\ell}^{\mathrm{Lap}}_{1b} - \widehat{\ell}^{\mathrm{Lap}}_{0b}\}$.
Set $p_b \leftarrow \Pr\{\chi^2_{q_b} \geq D_b\}$.
Set $\log \widehat{\mathrm{BF}}^{\mathrm{BIC}}_{10,b} \leftarrow (D_b - q_b \log M_g)/2$.

Algorithm 13 Inference comparison for the paired EC-count GLMM

Require: Paired counts $\boldsymbol{c}_{gip}$, maps A_{gp}, nested fixed-effect designs, mouse indices, likelihood family, and Rényi order $\alpha = 0.5$

Ensure: Null and alternative evidence approximations and coefficient estimates

1: **for all** null and alternative designs **do**
2: Fit the conditional posterior mode of $u_g \in \mathbb{R}^{M_g \times d_g}$ by damped Newton updates.
3: Optimize the Laplace marginal likelihood over B_g, $\log \sigma_g$, and optional $\log \kappa_g$.
4: **if** the likelihood is multinomial **then**
5: **repeat**
6: Update every latent EC-to-isoform responsibility from $A_{gpkt} \exp(E_Q \eta_{git})$.
7: Iterate every tilted coordinate a_{gip} to its softmax fixed point.
8: Update $B_g, \mu_g, \log s_g, \log \sigma_g$ against the bounded ELBO; damp the global update when the bound decreases.
9: **until** the ELBO gradient tolerance or iteration limit is reached
10: **end if**
11: Draw fixed antithetic standard-normal samples and reparameterize $u_g = \mu_g + s_g \odot \epsilon$.
12: Optimize the exact-likelihood Monte Carlo ELBO $\mathcal{L}_1$.
13: Warm-start from the ELBO fit and optimize $\mathcal{L}_{0.5}$; report its effective importance-sample size.
14: **end for**
15: Compare each method's null-to-alternative gain on its own objective; do not compare raw objective values across approximation families.

Algorithm 14 Block-specific paired EC-count GLMM cell-type test

Require: Block b, paired gene EC counts $\boldsymbol{c}_{gip}$, maps A_{gp}, condition design Z_b, cell-type design W_b, mouse indices, and R parametric-bootstrap replicates
Ensure: Empirically calibrated block p-value

```
Construct the Helmert basis $Q_b$ and isoform-to-path contrast matrix $H_b$; retain unmapped isoforms as likelihood nuisances.
Form the null fixed-effect tensor from unrestricted condition-by- isoform coefficients $Z_b \otimes I_{d_g}$ and cell-type nuisance columns $W_{bij}(N_b)_{:s}$.
Fit the null full-covariance posterior and variance parameters.
Append the $(J_b - 1)d_b$ columns $W_{bij}(H_b)_{:s}$, warm-start all null parameters, and fit the observed alternative.
Set $T_b \leftarrow 2(\mathcal{L}_{1b} - \mathcal{L}_{0b})$, using the same variational objective for both nested fits.
for $r = 1, \ldots, R$ do
    Draw mouse effects and optional pseudobulk effects from the fitted null; compute primer-specific EC probabilities.
    Generate multinomial or Dirichlet–multinomial EC counts while preserving every observed primer-specific pseudobulk total.
    Refit the null, warm-starting from the observed null, then append the block-effect columns and fit the nested alternative.
end for
Pool converged bootstrap statistics by likelihood, tested dimension, and depth stratum, excluding block $b$'s replicates.
Compute the finite-pool rank p-value and assess the same rank calculation on every null replicate before applying BH.
```

Algorithm 15 Matched semi-synthetic power benchmark

Require: Two-condition partitions, effect sizes γ, and replicate count R_{power}
Ensure: Asymptotic and null-calibrated power for each method

```
for all eligible blocks $b$ do
    Draw and fix a unit ILR direction $\boldsymbol{v}_b$.
    for $r = 1, \ldots, R_{\text{power}}$ do
        Permute condition labels among mice.
        for all nonzero effect sizes $\gamma$ do
            Set $\boldsymbol{y}_{bmc}^{(r,\gamma)} = \boldsymbol{y}_{bmc} + \gamma z_{bmc}^{(r)} \boldsymbol{v}_b$.
            Map shifted ILRs to compositions and effective path counts.
            Fit GEE, GLMM, clustered GLS, and Dirichlet–multinomial tests.
        end for
    end for
end for
Rank each alternative statistic against null statistics for its block and method, using the finite-permutation correction.
Report asymptotic and event-matched permutation rejection at 0.05.
```

Algorithm 16 Pairwise and dominant-path sensitivity tests

Require: Path estimates and conditional covariances from Algorithm 5
Ensure: Calibrated sensitivity tables

for all cell-type pairs **do**
  Retain M mice containing both types.
  Use $X_0 \in \mathbb{R}^{2M\times M}$ for mouse effects and $X_1 \in \mathbb{R}^{2M\times(M+1)}$ for mouse plus cell type.
  Permute the two labels within mouse and correct across all block-by-pair hypotheses.
end for
for all blocks b **do**
  Rank paths by $\sum_i N_{bi}^{\text{gene}}\psi_{bis}$ and select $\mathcal{S}_b$, the two largest, without labels.
  Construct selector E_b, two-path basis $Q_b^{(2)}$, and $\mathcal{R}_b = (Q_b^{(2)})' E_b' Q_b$.
  Set $V_{bi}^{(2)} = \mathcal{R}_b V_{bi} \mathcal{R}_b'$ and renormalize the two retained proportions.
  Run Algorithm 9 and record path-floor sensitivity sweeps.
end for

Algorithm 17 Joint condition tests

Require: Calibrated block-by-cell-type p-values or ILR responses
Ensure: Partition- or gene-level joint p-values

For ACAT, combine valid cell-type p-values within a canonical partition; optionally combine partition p-values within a gene.
As a negative control, fit one shared condition effect while treating cell-type pseudobulks as independent.
For clustered GLS, construct diagonal covariance blocks $V_{bmc} + (\tau^2_{\text{mouse}} + \tau^2_{\text{resid}})I_{d_b}$ and same-mouse off-diagonal blocks $\tau^2_{\text{mouse}} I_{d_b}$.
Estimate τ^2_{mouse} and τ^2_{resid} by REML under the null.
Fit the shared condition coefficient by GLS.
for all mouse-level condition permutations **do**
  Apply one permuted condition to all cell types from that mouse and refit the model.
end for
Apply BH to the canonical-partition family; keep negative-control and calibrated results separate.

C Supporting Implementation Details

Splice-block construction

GTF exons are represented by zero-based, half-open intervals. Exon boundaries define atomic intervals; intervals exonic in every annotated spliced isoform are merged into constitutive anchors. Consecutive anchors define a block, and an isoform path is its ordered exon-interval chain between them. Terminal pseudo-anchors include alternative first and last regions. A gene without a shared constitutive region receives one whole-gene block.

Blocks with one path are discarded. Simple two-path blocks include skipped exons, mutually exclusive exons, alternative donors and acceptors, and retained introns. The implementation also supports nested and multi-path blocks. Precursor isoforms remain nuisance isoforms in the EC model but are not assigned to spliced paths.

Different genomic blocks can induce exactly the same partition of annotated isoforms, for example when several exons always occur together. Such blocks have identical statistical responses and must not count as repeated evidence. A canonical partition is the unordered collection of isoform sets assigned to paths. Equivalent blocks are collapsed before FDR or joint testing; all member interval identifiers remain in the output.

Pseudobulks and local fitting

Cell isoform proportions from the hierarchical decoder are weighted by cell UMI totals and aggregated by mouse, condition, and cell type. Pseudobulks must contain at least 20 cells and 10^5 total UMIs. A block is fit when its gene has at least 25 gene-assigned UMIs in at least two qualifying pseudobulks. Only ECs whose compatible isoforms map uniquely to that gene enter the local likelihood.

Bounded L-BFGS optimizes the paired multinomial likelihood, with $\|\boldsymbol{\delta}_{bi}\|_\infty \leq 20$ to prevent floating-point underflow, where $\|\cdot\|_\infty$ is the maximum absolute component. Two-path fits are initialized from a log-ratio grid because the paired-primer normalizers can make the objective non-concave.

For the cell-resolved variant, sparse cell-by-EC matrices are retained and one local isoform matrix is decoded at a time. The optimizer dimension is still only the number of paths minus one. Cell-specific Fisher contributions are accumulated analytically, so the method does not fit one free splicing parameter per cell.

Identifiability and reliability

Finite Fisher covariance is not sufficient near a simplex boundary. Parametric bootstrap identified two reliability requirements:

$$\lambda_{\max}(V_{bi}) \leq 1/8 \quad \text{and} \quad \min_s \widehat{\psi}_{bis} \geq 0.15.$$

Thus every tested log-ratio direction has effective information of at least eight and every fitted path has at least 15% usage. All estimates are written, together with convergence, identifiability, and reliability flags; only reliable estimates enter GLS. Here $\lambda_{\max}(V_{bi}) \in \mathbb{R}_+$ is the largest eigenvalue of the $d_b \times d_b$ covariance, and $\widehat{\psi}_{bis} \in [0,1]$ is path s's fitted proportion.

D Supporting Results and Benchmarks

Condition-test scope

The primary analysis used the 25-gene-UMI cutoff and blocks with at most eight represented paths. It fit 10,100 blocks and 359,713 pseudobulk-by-block path estimates; 359,711 fits converged. After reliability filtering, there were 829 conditional and 506 profile tests.

These 1,335 fits represent 829 distinct block-by-cell-type hypotheses, covering 527 distinct splice blocks in eight cell types. The profile analysis is available for a subset of the conditional hypotheses. The possible conditions were 5xFAD, 5xFIRE, 5xFIRE PBS, 5xFIRE transplant, FIRE, and WT. For the 829 conditional hypotheses, 275 compared two conditions, 155 compared three, 153 compared four, 130 compared five, and 116 compared all six. The included set varies because every condition in a test must have at least three independent mouse pseudobulks for that block and cell type.

Condition-test results

No block was significant at 5% FDR in either covariance analysis. At the unadjusted $p < 0.05$ level:

- the conditional analysis identified 24 block-by-cell-type hypotheses, covering 21 distinct blocks;
- the profile analysis identified 20 block-by-cell-type hypotheses, covering 17 distinct blocks;
- 20 hypotheses were nominally significant in both analyses.

The union is 24 nominal block-by-cell-type hypotheses covering 21 blocks, but the corrected number of significant hypotheses and blocks is zero. Thus there is no FDR-significant evidence for condition-dependent splicing within cell type. The two covariance variants should not be added as if they were separate discoveries.

Formal mouse-clustered models

The shared-condition analysis was repeated with the multinomial-logit GLMM and Liang–Zeger GEE defined above. Clustered Gaussian ILR GLS and the unclustered Dirichlet–multinomial were refit to the same canonical partitions as comparators. Each block used 50 synchronized mouse-level null permutations. For 88 two-condition blocks, 50 semi-synthetic replicates were generated at each ILR effect size; 4,369 of 4,400 replicate designs had full rank. "Perm. nominal" below is the number of observed partitions with an event-matched permutation p-value at most 0.05. Table 11 summarizes the complete analysis.

Table 11: Calibration and power comparison. Null rejection uses asymptotic p-values on permuted labels; power uses event-matched permutation ranks.

Method	Tests	Conv.	Null $< .05$	Perm. nominal	Power .25	Power .50
Clustered GLS	359	359	0.025	17	0.171	0.322
Dirichlet–multinomial	359	359	0.138	18	0.188	0.364
Liang–Zeger GEE	359	355	0.415	8	0.139	0.268
Laplace GLMM	359	359	0.108	19	0.199	0.390

The canonical GEE sandwich reference is unusable at the available mouse counts: 41.5% of permuted-null p-values were below 0.05, and its apparent 116 asymptotic FDR discoveries are artifacts of this inflation. The GLMM reference was closer but still anti-conservative at 10.8%, while clustered GLS was conservative at 2.5%. Event-matched permutation calibration is therefore required for all comparisons. On that scale, the GLMM had the highest power, improving over clustered GLS by 2.8 and 6.9 percentage points at ILR effects 0.25 and 0.5. GEE was less powerful than clustered GLS after calibration. No method produced a permutation-calibrated 5% FDR discovery in the observed condition analysis.

EC-count GLMM inference benchmark

The EC-count methods were first compared in 96 independent simulations per likelihood and effect size. Each replicate had 12 mice, two observations per mouse, two isoforms, two distinct three-EC primer maps, 150 molecules per primer and observation, and mouse random-effect standard deviation 0.4. Dirichlet–multinomial simulations used concentration 30. The null threshold for each method was its own 95th percentile evidence gain at effect zero; power was evaluated at an isoform logit effect of 0.5. This calibration is required because Laplace likelihoods, tilted ELBOs, and Rényi bounds are not numerically interchangeable objectives.

Table 12: Ambiguous-EC simulation comparison. RMSE is for the condition logit coefficient. Time is median seconds per nested null/alternative fit at effect 0.5. ESS is the median alternative mouse-level importance ESS out of 128.

Likelihood	Approximation	Null rejection	Power	RMSE	Time	ESS
Multinomial	Laplace	0.052	0.542	0.223	9.87	–
Multinomial	Tilted ELBO	0.052	0.542	0.223	1.87	–
Multinomial	Rényi $\alpha = 0.5$	0.052	0.542	0.223	1.58	127.67
Dirichlet–multinomial	Laplace	0.052	0.417	0.236	21.41	–
Dirichlet–multinomial	Monte Carlo ELBO	0.052	0.417	0.236	3.06	127.51
Dirichlet–multinomial	Rényi $\alpha = 0.5$	0.052	0.417	0.236	3.35	127.66

Within each likelihood family, all three approximations had the same empirically calibrated power to the reported precision, and their coefficient bias differed by less than 2×10^{-4}. The tilted bound's null cutoff was 4.70, compared with 4.60 for multinomial Laplace and Rényi VI, but this small scale difference did not change rejection. Variational fitting was five- to seven-fold faster than Laplace. A single global importance weight over all mice had ESS 1/128 and collapsed the fitted random-effect variance; the mouse-factorized Rényi objective corrected this failure and produced ESS above 127.5/128 for both likelihoods.

A second simulation used three isoforms and otherwise retained the same 96 null and 96 effect replicates. Full-covariance and diagonal VI were nearly indistinguishable. Tilted evidence gains had Spearman correlation 0.9999 and median absolute difference 0.073 between posterior families; Rényi gains had correlation 0.99996 and median absolute difference 0.044. Every VI variant had calibrated power 0.438 and condition-coefficient RMSE 0.237–0.238. The full-covariance tilted fit took a median 2.43 seconds per nested effect fit versus 2.00 seconds for diagonal tilted VI. Thus, in a controlled multi-isoform setting, omitted posterior covariance was not a material source of error.

The same conclusion held for the direct EC Dirichlet–multinomial. In 96 three-isoform null and effect simulations, diagonal and full ELBO gains had correlation 0.9997 and median absolute difference 0.150; diagonal and full Rényi gains had correlation 0.9999 and median difference 0.092. Both posterior families had calibrated power 0.219. In the 65-gene real-data analysis, diagonal and full ELBO gains had correlations 0.998 for condition and 0.9998 for cell type, with median absolute differences 0.0035 and 0.0050. Fixed-draw Rényi optimization was again less stable than ELBO optimization.

The genome-wide paired-primer comparison retained 65 genes with mean depth at least 100 EC UMIs per pseudobulk; this was 65 of 66 depth-eligible genes, with one 3,905-EC outlier excluded by the 128-EC computational cap. All retained genes had at most six EC-supported isoforms. The authoritative run initialized each Gaussian variational posterior from the corresponding Laplace means and covariance diagonal, then initialized Rényi VI from the ELBO fit. Across the 65 condition models, convergence was 98.5% for both Laplace likelihoods, 95.4% for both ELBO variants, 92.3% for multinomial Rényi VI, and 98.5% for Dirichlet–multinomial Rényi VI. Cell-type convergence ranged from 93.8% to 100%. There were no fit exceptions.

Approximation agreement depended on isoform dimension. Among the 38 two-isoform genes, multinomial Laplace and Rényi evidence gains had Spearman correlation 0.974 for condition and 0.9996 for cell type; Laplace and tilted ELBO correlations were 0.969 and 0.9995. All 16 two-isoform condition genes above the method-specific nominal threshold under multinomial Laplace were also above it under Rényi VI. For the 20 three-isoform genes, the analogous gain correlations fell to 0.49 and 0.59. The mean-field posterior has only a diagonal covariance across isoform random effects, whereas Laplace retains their within-mouse posterior covariance. The dimension-dependent divergence therefore motivated a full-covariance extension, but by itself did not distinguish covariance error from objective or optimizer differences. One-dimensional simulation thresholds were not applied to multi-isoform or cell-type omnibus tests.

The tilted surrogate must also be separated from objective evaluation. For two-isoform condition models, median nested gains were 5.54 for the tilted training bound, 4.95 for an exact-likelihood Monte Carlo ELBO evaluated at the same posterior, 4.93 for its Rényi evaluation, 4.93 for Laplace, and 4.95 for separately optimized Rényi VI. For cell type, the corresponding medians were 89.06, 74.38, 74.36, 72.12, and 74.36. Exact ELBO and Rényi gains on the same posterior had Spearman correlations 0.998 and 1.000 for condition and cell type. The tilted bound becomes looser under the larger alternative and therefore overstates nested evidence gains; it is retained as a fitting surrogate, not used as a likelihood-ratio statistic.

Full-covariance results did not support posterior covariance as the cause of the earlier multi-isoform discrepancy. On the real condition analysis, fresh-sample ELBO gains from diagonal and full tilted posteriors had Spearman correlation 0.991 across 65 genes and median absolute difference 0.002. Among the 20 three-isoform genes, the correlation was 0.938 and median difference was 0.024. The corresponding cell-type values were 1.000 overall and within the three-isoform subset, with median three-isoform difference 0.023. These results, together with the three-isoform simulation, point instead to differences in the Laplace objective or its outer optimization. Separately optimizing a fixed-draw full-covariance Rényi objective was less stable than fitting the tilted objective and evaluating both exact objectives with fresh draws; the latter is the reported comparison.

The observation-level logistic-normal model was validated with 96 null and 96 effect simulations using three isoforms, true residual standard deviation 0.6, and the same ambiguous paired-primer maps. Tilted full-covariance VI estimated median residual standard deviation 0.559 under the effect and had calibrated power 0.354. Dirichlet–multinomial Laplace had power 0.365, while the misspecified ordinary multinomial had power 0.313. Logistic-normal Laplace had power 0.146 despite estimating residual scale 0.472, indicating that its high-dimensional marginal approximation was not competitive with the tilted fit.

In the real data, tilted logistic-normal fits converged under both nested models for 80.0% of condition genes and 86.2% of cell-type genes. The alternative residual standard deviation had median 0.040 for condition and 0.029 for cell type; the respective interquartile ranges were 0.007–0.239 and 0.006–0.234. Relative to the multinomial full posterior, fresh ELBO condition gains changed by a median absolute 0.335 and retained rank correlation 0.809. Cell-type gains changed by 10.68 and retained correlation 0.959, consistent with extra-multinomial variation having more impact on the larger cell-type omnibus statistic. Because residual-scale estimates often approached zero and convergence was weaker, this model is retained as an overdispersion sensitivity analysis rather than replacing the primary test.

The initial block-specific EC analysis inherited 406 coverage-eligible partitions from the path-count analysis and tested 399 after EC identifiability filtering. This was unnecessarily restrictive because it required reliable path estimates before fitting the EC likelihood. The genome-wide EC screen instead includes one EC count vector for every pseudobulk–primer combination in the likelihood, retains cell types represented in at least three mice, and requires at least two represented cell types. The strict rule additionally requires at least 25 gene UMIs and 30 covered pseudobulks; the permissive rule requires at least 10 gene UMIs. These rules retain 1,926 and 5,650 blocks, respectively. Five parametric-bootstrap replicates per block provide pooled null distributions matched by tested dimension. Table 13 reports the calibrated tests.

Table 13: Parametric-bootstrap-calibrated cell-type tests using gene EC counts. The depth column gives the range of null rejection rates across independent gene-depth quartiles. MN is multinomial, LN is logistic-normal multinomial, and DM is Dirichlet–multinomial.

Coverage	Likelihood	Tests	Conv.	Nominal	FDR	Null < .05	Depth range
Strict	MN	1,926	1.000	415	174	0.0496	0.0480–0.0505
Strict	LN	1,926	0.991	377	157	0.0498	0.0439–0.0541
Strict	DM	1,926	0.988	145	0	0.0491	0.0454–0.0582
Permissive	MN	5,650	0.999	997	294	0.0497	0.0475–0.0519
Permissive	LN	5,650	0.984	974	193	0.0497	0.0428–0.0541
Permissive	DM	5,650	0.989	421	0	0.0498	0.0456–0.0516

The permissive rule increases absolute discoveries from 174 to 294 for MN and from 157 to 193 for LN, while testing 2.9 times as many blocks. MN and LN share 152 strict and 192 permissive discoveries. DM has no genome-wide FDR discoveries under either rule despite acceptable calibration. On the 1,926 shared hypotheses, applying BH only to that common family gives 316 MN, 265 LN, and 40 DM discoveries under the permissive fits, versus 174, 157, and zero under the strict fits. Thus lower-count pseudobulks add signal, but much of that gain is spent on the larger genome-wide hypothesis family. The corresponding strict-versus-permissive statistic rank correlations are 0.781, 0.764, and 0.760, so changing the cutoff also changes individual fits rather than only the BH denominator.

Expanded EC likelihood and coverage comparison

The final analysis uses one likelihood and testing construction for both biological contrasts: the logistic-normal multinomial block EC GLMM. For the cell-type test, condition indicators are nuisance fixed effects and the alternative adds block-path-by-cell-type interactions. For the condition test, models are fit separately within each cell type; the null has an intercept fixed effect and the alternative adds block-path-by-condition interactions. Both versions retain a mouse random effect and paired primer-specific EC likelihoods. Every retained condition must be represented by at least three mice, and every tested pseudobulk must have at least 10 modeled gene UMIs.

The annotation-level block equivalence relation is insufficient after EC screening: removing unsupported isoforms can make two distinct annotated blocks induce exactly the same partition of the remaining isoforms. The final hypothesis family therefore retains one representative for each tuple of gene, EC-supported isoforms, canonical block-path partition, covered observations, and tested factor levels. This reduces the permissive cell-type family from 5,650 to 4,825 tests and the condition family from 17,685 to 14,833 block-by-cell-type tests over 5,608 representative blocks. This collapse is performed before calibration and BH correction.

Table 14: The unified, nonredundant logistic-normal block EC GLMM after cross-fitted tail calibration. Condition tests are within cell type. Robust calls are significant at every tail threshold from 80% through 95%.

Contrast	Tests	Conv.	Nominal	FDR	Robust	Null < .05	Null < .01	Null < .001
Cell type	4,825	0.999	771	252	238	0.04999	0.00999	0.00099
Condition	14,833	1.000	1,397	0	0	0.05001	0.00999	0.00100

The initial empirical calibration compared each test only with its degrees-of-freedom-specific bootstrap pool. After supported-partition deduplication, its minimum attainable p-values were approximately 4.5×10^{-4}, whereas BH requires resolution near 10^{-5} at the first ranks. Its zero logistic-normal calls were therefore a bootstrap-tail resolution artifact, not disappearance of the cell-type signal.

The final calibration fits a generalized Pareto peaks-over-threshold model within each degrees-of-freedom stratum. Fits are cross-fitted by holding out whole block tests. The resulting observed and held-out-null tail scores are then recalibrated by a global leave-block-out empirical null CDF. This second stage corrects residual tail-model bias and gives sufficient pooled resolution for genome-wide BH. At a 90% threshold, multinomial, logistic-normal multinomial, and Dirichlet–multinomial EC tests identify 272, 252, and 59 nonredundant cell-type partitions. Logistic-normal thresholds of 80%, 85%, 90%, and 95% give 242, 240, 252, and 245 calls; 238 occur in all four sets, 231 of those are multinomial calls, and 45 are called by all three likelihoods.

Condition is less stable. The same four thresholds give zero, 28, zero, and zero calls, with no partition in all four sets; at the 90% threshold the minimum FDR is 0.0545. The final conclusion is therefore strong calibrated cell-type DS, but no robust condition DS. The earlier 193 logistic-normal calls in Table 13 used annotation-level duplicates and under-resolved empirical tails and are superseded by the 238 robust set.

Cell-type effects within mouse

The initial Gaussian analysis tested 369 conditional and 218 profile partitions. It found 22 and five FDR-significant partitions, respectively, with all profile discoveries contained in the conditional set. After canonicalizing annotation intervals that encode the same isoform-to-path partition, these 22 intervals represented 14 distinct partitions.

The compositional analysis used converged, conditionally identifiable estimates and required every modeled path proportion to be at least 5%. This produced 646 nonredundant tests. A minimum of 30 pseudobulk observations per partition, chosen as a label-effect-independent coverage requirement, retained 406 hypotheses for FDR correction. Among these, 149 had exact permutation $p < 0.05$, and **101 splice-path partitions in 95 genes passed 5% FDR**. Thirteen of the 14 nonredundant Gaussian discoveries were recovered; the remaining partition had exact $p = 0.032$ but FDR 0.098. Significant compositional results had median maximum cell-type logit effect 0.84 and median effective path count 21.7.

The observation-count threshold is not a narrow boundary effect: thresholds from 29 through 32 pseudobulks each gave 101 discoveries, while 33 gave 99. All 646 observed fits converged. The median number of valid permutations was 500; the minimum was 476 after excluding failed permutation refits. Asymptotic Dirichlet–multinomial p-values were anti-conservative, with 15.5% of permutation-null values below 0.05, which is why only exact per-event permutation ranks are used for the primary result.

Two sensitivity analyses did not improve the event-level result. Testing 3,708 individual cell-type pairs gave 97 FDR-significant pair hypotheses but only 32 distinct partitions after correction across all pairs. Restricting each block to its two molecule-weighted dominant paths increased coverage; with a 1% path floor it gave 88 discoveries, and removing the floor gave 91, under a cross-fitted empirical null. These analyses support the full-path exact omnibus test as the primary cell-type analysis.

Allowing the alternative to re-estimate κ also left the result unchanged: it found the same 101 partitions in the same 95 genes, with no event entering or leaving the 5% FDR set. Exact p-values from fixed- and free-concentration fits had Spearman correlation 0.9995, and both had 148 nominal events. The null optimum was multinomial for 499 of 646 tests, and the alternative optimum was multinomial for 580. Among 66 tests with finite concentration in both models, the median alternative/null concentration ratio was 1.54, consistent with cell-type mean effects explaining some apparent overdispersion. Exact permutation calibration absorbs this nuisance-parameter flexibility, so freeing κ provides no net power gain here.

Calibration

Across the 25-UMI analysis, 2.7% of permuted-label Wald p-values were below 0.05 for both conditional and profile covariance, indicating conservative null calibration. A parametric bootstrap of 3,000 reliable local fits gave mean squared standardized error 1.10 and 94.5% coverage for nominal 95% intervals. Independent Gaussian simulations of REML–GLS gave type-I error between 2.7% and 5.0% across the tested biological-variance settings.

Numerical tests additionally verify skipped-exon block construction, identifiable covariance after collapsing indistinguishable isoforms, rejection of information-null path contrasts, the analytic binomial Fisher variance, and recovery of imposed GLS effects.

Coverage-cutoff sensitivity

UMIs	blocks	tests	conditional null $p < .05$	profile null $p < .05$	bootstrap coverage
25	10100	1335	0.027	0.027	0.945
50	5896	755	0.031	0.030	0.936
100	2820	283	0.029	0.025	0.946
200	1034	69	0.028	0.026	0.948
500	195	10	0.030	0.050	0.946

No cutoff produced an FDR-significant block. The 500-UMI calibration estimate is based on only 100 permutation tests.

For reliable two-condition, two-path conditional tests, mean asymptotic power for a 0.5 ILR effect was 0.49, 0.51, 0.44, 0.56, and 0.59 at increasing cutoffs. The final two values are selected from only eight and two eligible tests. Among 103 hypotheses shared by the 25- and 50-UMI analyses, power was 0.453 and 0.449. Lower coverage cutoffs therefore increase the number of testable blocks without degrading calibrated precision among shared tests. The 25-UMI cutoff is the default.

Does cell resolution increase condition-test power?

The cell-resolved likelihood was run on the 527 blocks testable in the 25-UMI primary analysis. All 41,242 block-by-pseudobulk fits converged; 21,624 passed the reliability rules, compared with 21,010 pseudobulk conditional estimates on the same blocks. This produced 836 condition tests, compared with 829 for pseudobulk fitting. No cell-resolved test passed 5% FDR. Condition-label permutations were conservative, with 2.49% of null p-values below 0.05.

An exact matched comparison used the same reliable samples for both methods in 807 block-by-cell-type tests. Cell resolution reduced the median local covariance trace by 1.6%, and the median path-response RMS difference was 0.033 ILR units. P-values were strongly concordant (Spearman rank correlation 0.958); 24 pseudobulk and 25 cell-resolved tests were nominally significant.

For the 259 matched two-condition, two-path tests, mean asymptotic power for a 0.5 ILR effect was 0.504 for both methods. The median ratio of cell-resolved to pseudobulk condition-coefficient standard error was 0.995. Retaining cell likelihood terms therefore did not materially increase power in this dataset. The aggregated EC counts and decoder baseline preserve nearly all relevant local information within a cell type, while residual between-mouse variation dominates the small reduction in inferential variance. Pseudobulk fitting remains the default; the cell-resolved method is available for datasets with stronger within-pseudobulk isoform heterogeneity.

Compositional and joint-test assessment

Collapsing equivalent isoform partitions reduced the 829 conditional block-by-cell-type fits to 505 nonredundant tests over 317 partitions. The median effective path count was 32 molecules, or 62% of gene UMIs; only 0.6% of estimates reached the gene-UMI cap.

On the same 505 tests, conditional GLS had 12 nominal results and the Dirichlet–multinomial had 18. Neither had a 5% FDR discovery. The Dirichlet–multinomial was conservatively calibrated: 3.76% of 10,100 permuted-label p-values were below 0.05. In contrast, pure multinomial regression gave 197 nominal and 155 FDR results, but 39.1% of its permutation p-values were below 0.05. UMI sampling alone therefore cannot represent the large between-mouse compositional variation. Output tables mark pure multinomial and naive shared-effect fits as invalid for inference while retaining them as calibration diagnostics.

ACAT across cell types produced 8 nominal GLS and 12 nominal Dirichlet–multinomial partitions among 317 tests, with no FDR discoveries. Combining distinct partitions within genes gave 7 and 11 nominal genes among 276 tests, again with no FDR discoveries.

Naive shared-effect pooling illustrates the pseudoreplication risk. It gave 43 nominal and one FDR Dirichlet–multinomial partition, but synchronized mouse permutations rejected 13.2% of nulls at 0.05. The apparent discovery had asymptotic $p = 6.4 \times 10^{-5}$ but permutation $p = 0.143$. Naive shared GLS was also inflated, with 8.4% null rejection.

The mouse-clustered shared GLS restored conservative calibration: 2.19% of 7,169 synchronized permutation p-values were below 0.05. It produced 11 nominal partitions among 359 tests and no FDR discoveries; its smallest p-value was 0.0126 with FDR 0.86. Gene-level ACAT gave 9 nominal genes among 313 and no FDR discoveries. Thus the calibrated gain is limited to the increase from 12 to 18 nominal within-cell-type tests under the Dirichlet–multinomial. None of the evaluated strategies creates a genome-wide condition discovery in this dataset.

E Limitations and Extensions

The profile covariance is conditional on the fitted genome-wide decoder. Uncertainty in shared decoder parameters is expected to be small for common paths but may matter for rare genes or cell types. A mouse-level jackknife of the hierarchical fit is the preferred extension.

The current biological variance is isotropic in log-ratio space. Estimating a full covariance separately for every block would be unstable with few mice. A later empirical-Bayes model could shrink path-specific biological covariances across genes.